



GE Medical Systems

gehealthcare.com

Technical Publication

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Revision 7

GE Medical Systems LightSpeed 1.X (QXi) - Functional Interconnects

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

LANGUAGE (CONTINUED)

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

LANGUAGE (CONTINUED)

警告

このサービスマニュアルには英語版しかありません。
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本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

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Revision History

Revision	Date	Reason for change
0	March 3, 1999	Initial Release for QXi
1	March 15, 1999	Changed drawing line weights to 0.25 point (CQA991635)
2	August 23, 1999	Drawings 2126185, 2126225, 2126219, & 2126245 updated.
3	March 15, 2000	CQA 1000177, Drawing 2126219 and duplicate drawing 2217013 removed. Drawings 2269964 and 2269967 added.
4	October 26, 2000	Drawings 2217013 and 2269967 updated.
5	October 26, 2001	CQA 1012427: Removed/Replaced incorrect pages.
6	February 12, 2002	CQA 1020694: Added drawing 2151953 Rev. 0 Sheets 1 and 2
7	August 17, 2004	Drawing 2123540 updated.

Preface

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task.

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**



**DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT**

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



**WARNING
ROTATING
EQUIPMENT
BARE WIRES**

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.



**NOTICE
Equipment
Damage
Possible**

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could cause harm.

ELECTRICAL



MECHANICAL



RADIATION



LASER



HEAT



PINCH



Some others make you aware of specific procedures that should be followed.

**AVOID STATIC
ELECTRICITY**



TAG AND LOCK OUT



**WEAR EYE
PROTECTION**



Section 2.0 Conventions Used For General Information

2.1 Document Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

2.2 Page Layout

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output and Input

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application.

Example: *This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>*

Example: **This paragraph denotes fixed input. It's typed input that will not vary from application to application.**

Example: ***This paragraph denotes computer input that can vary from application to application. Variable input will normally be found placed between greater than and lesser than operators. For example: <variable_input>***

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Whereas the computer **MENU** button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

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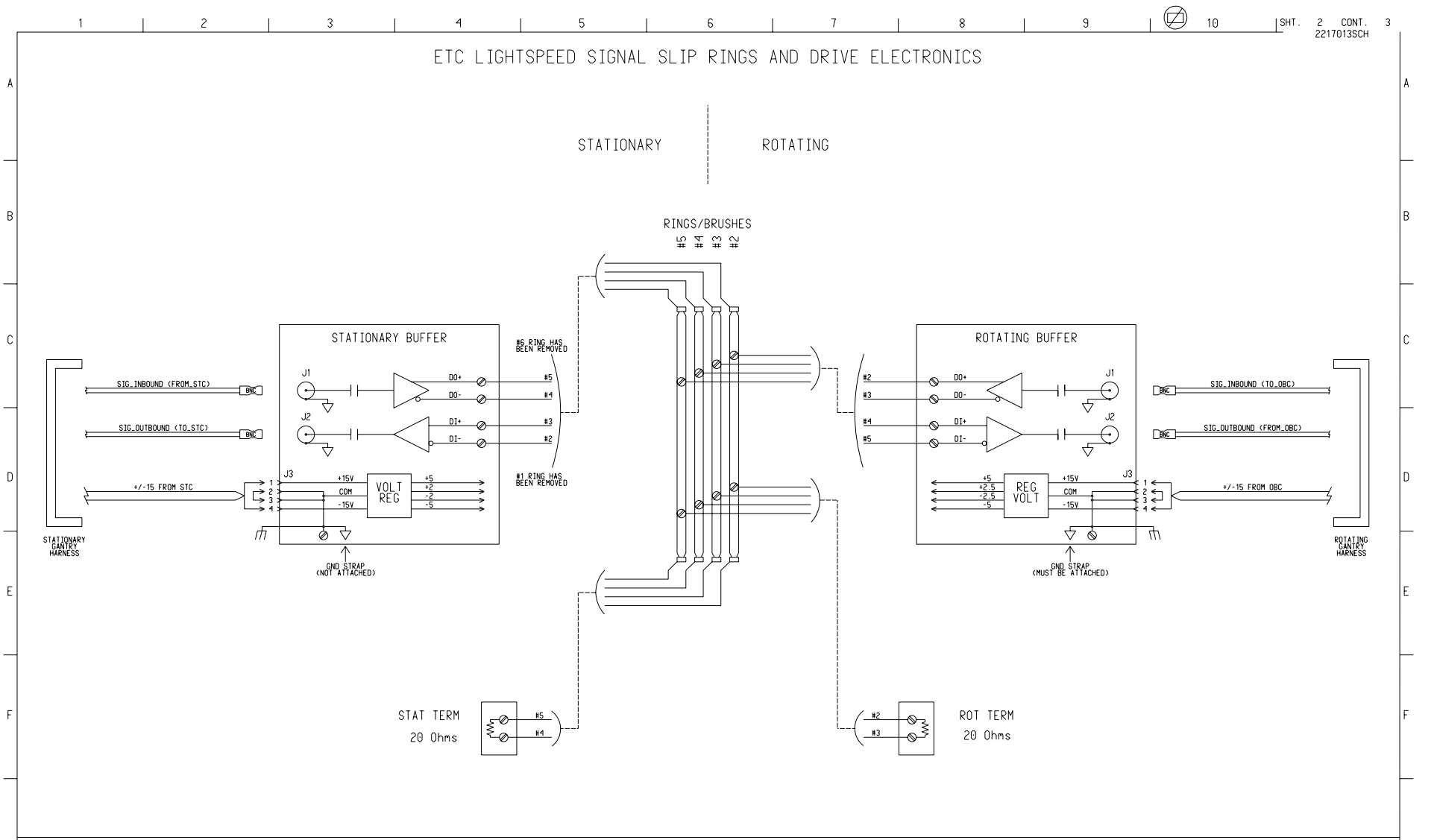
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Chapter 1

System Communications



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Figure 1-2 2217013 Sheet 2, Rev. 0

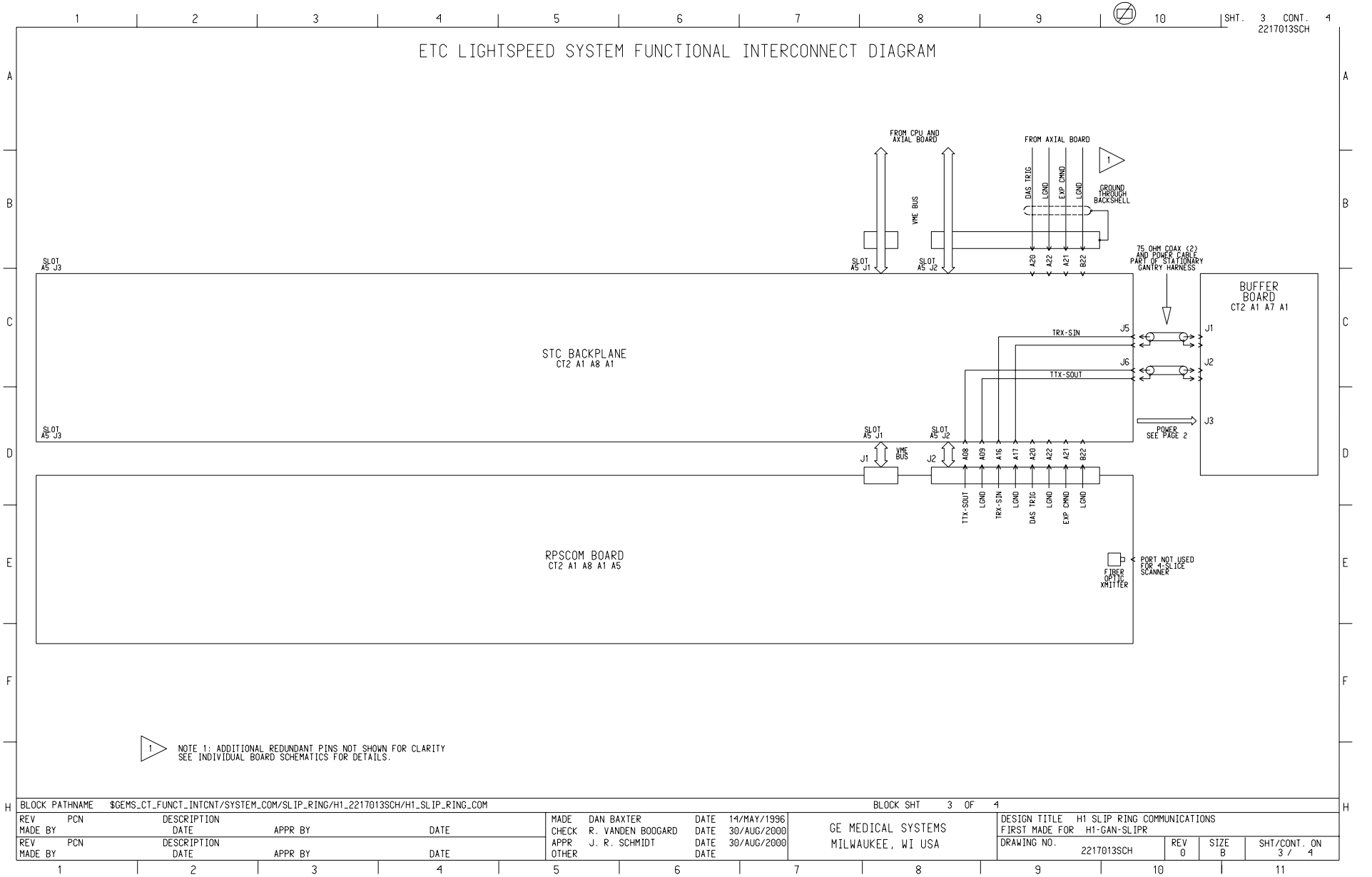
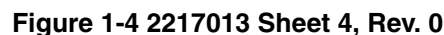


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Section 2.0 - Gantry Slip Ring Communications (S/A) - 2269967

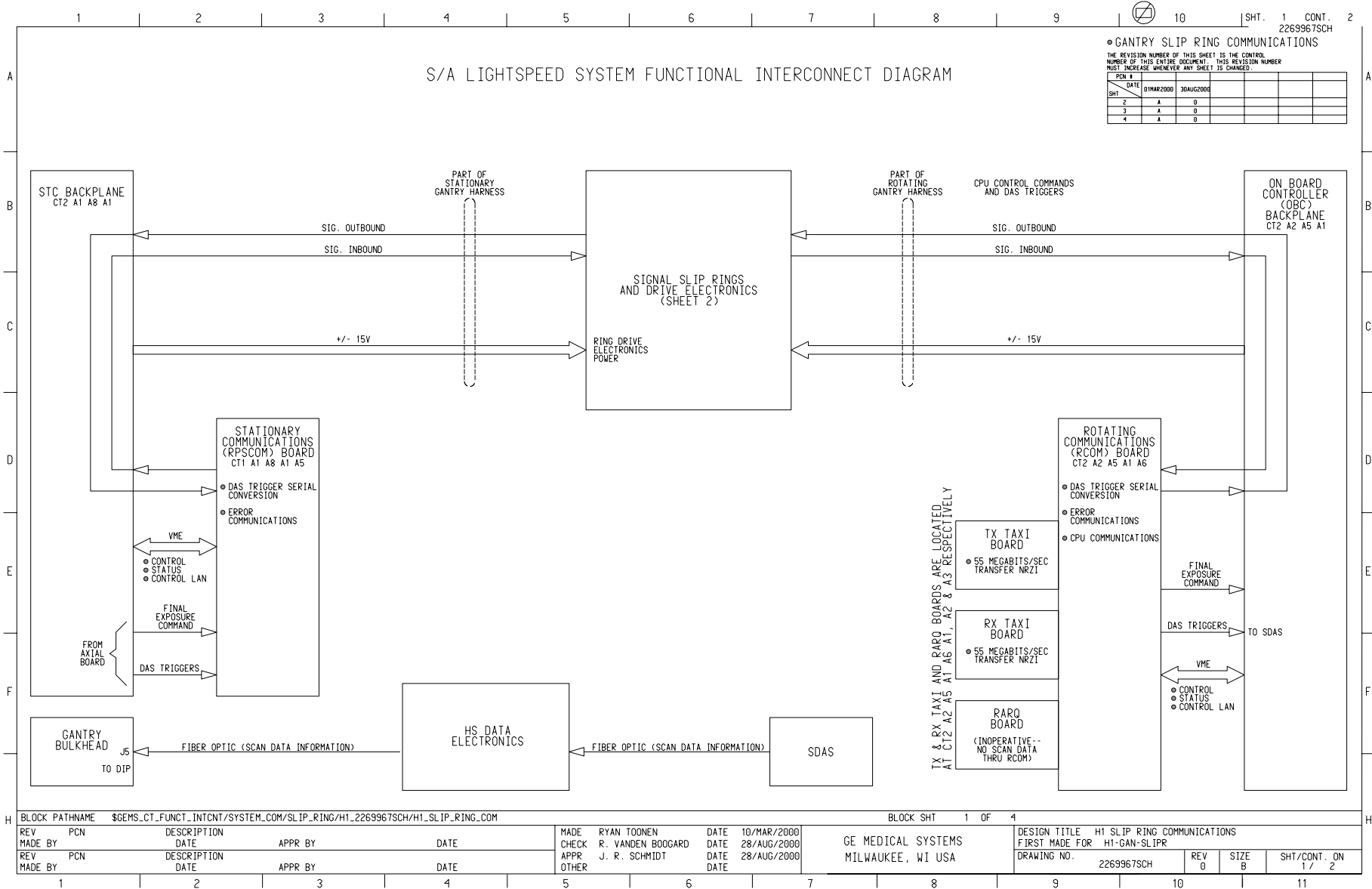
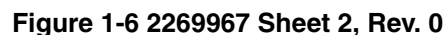


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Section 2.0 - Gantry Slip Ring Communications (S/A) - 2269967

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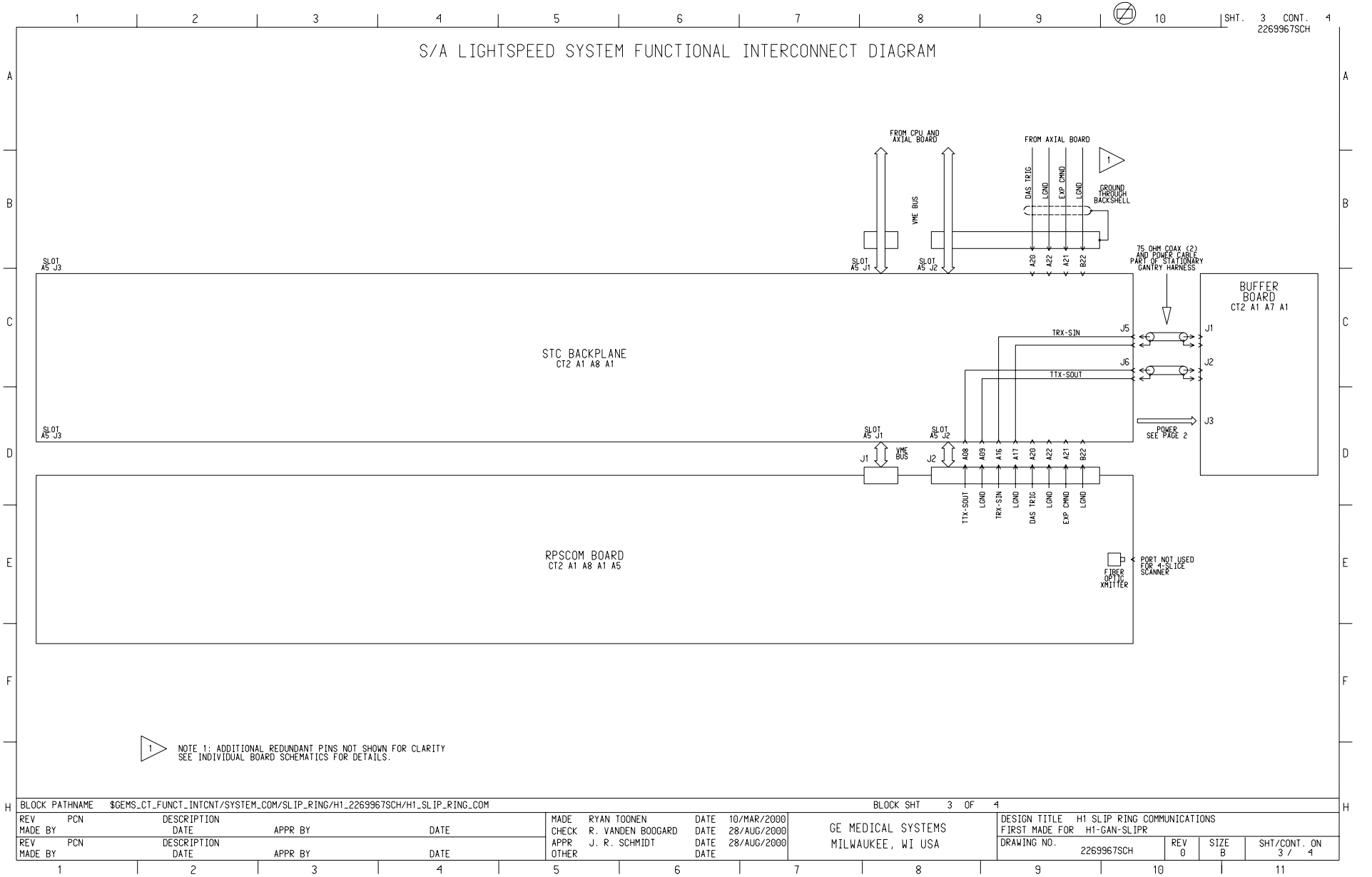
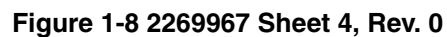
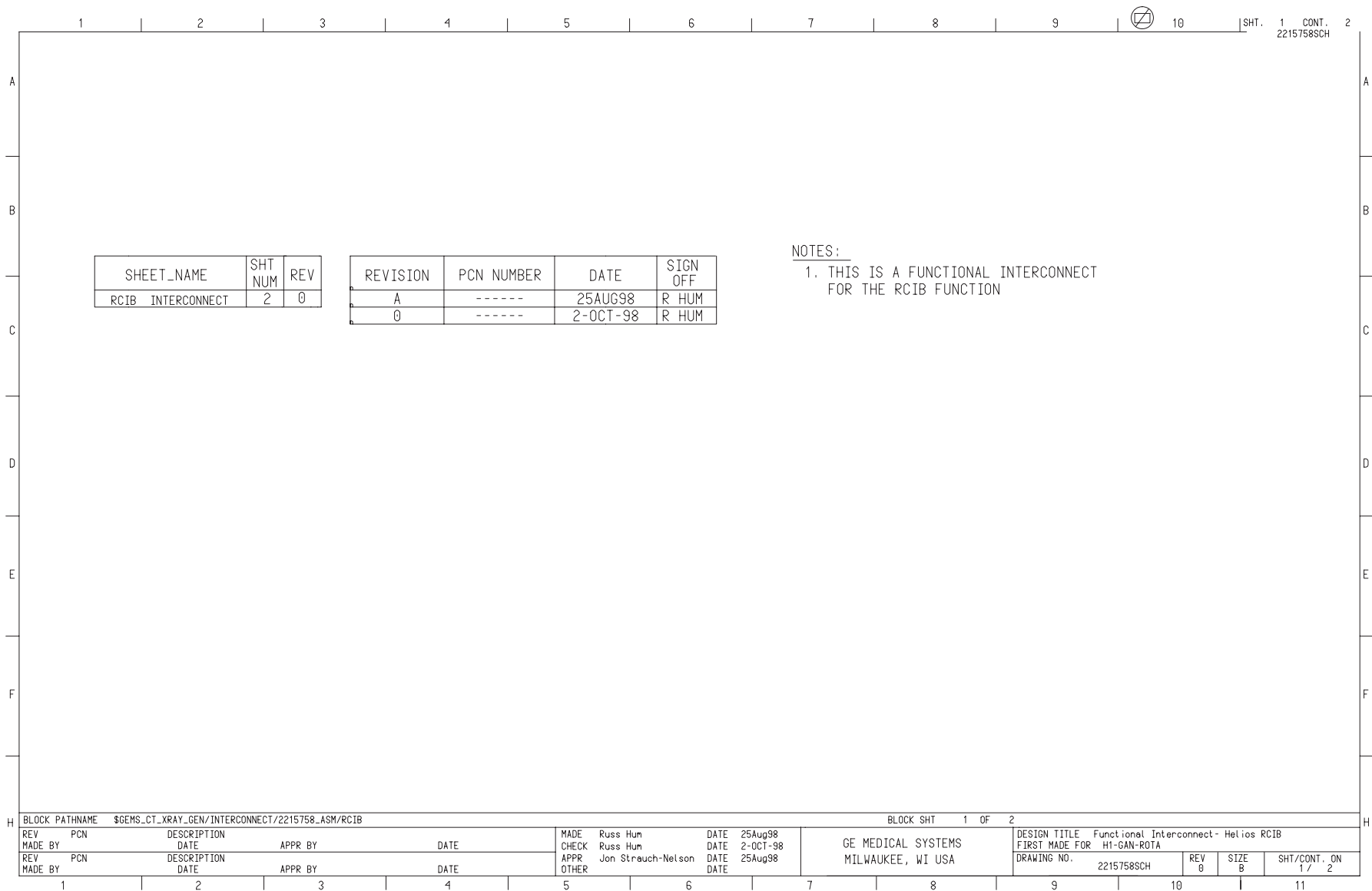


Figure 1-7 2269967 Sheet 3, Rev. 0



Section 3.0 - Control Network with RCIB/GCAN - 2215758



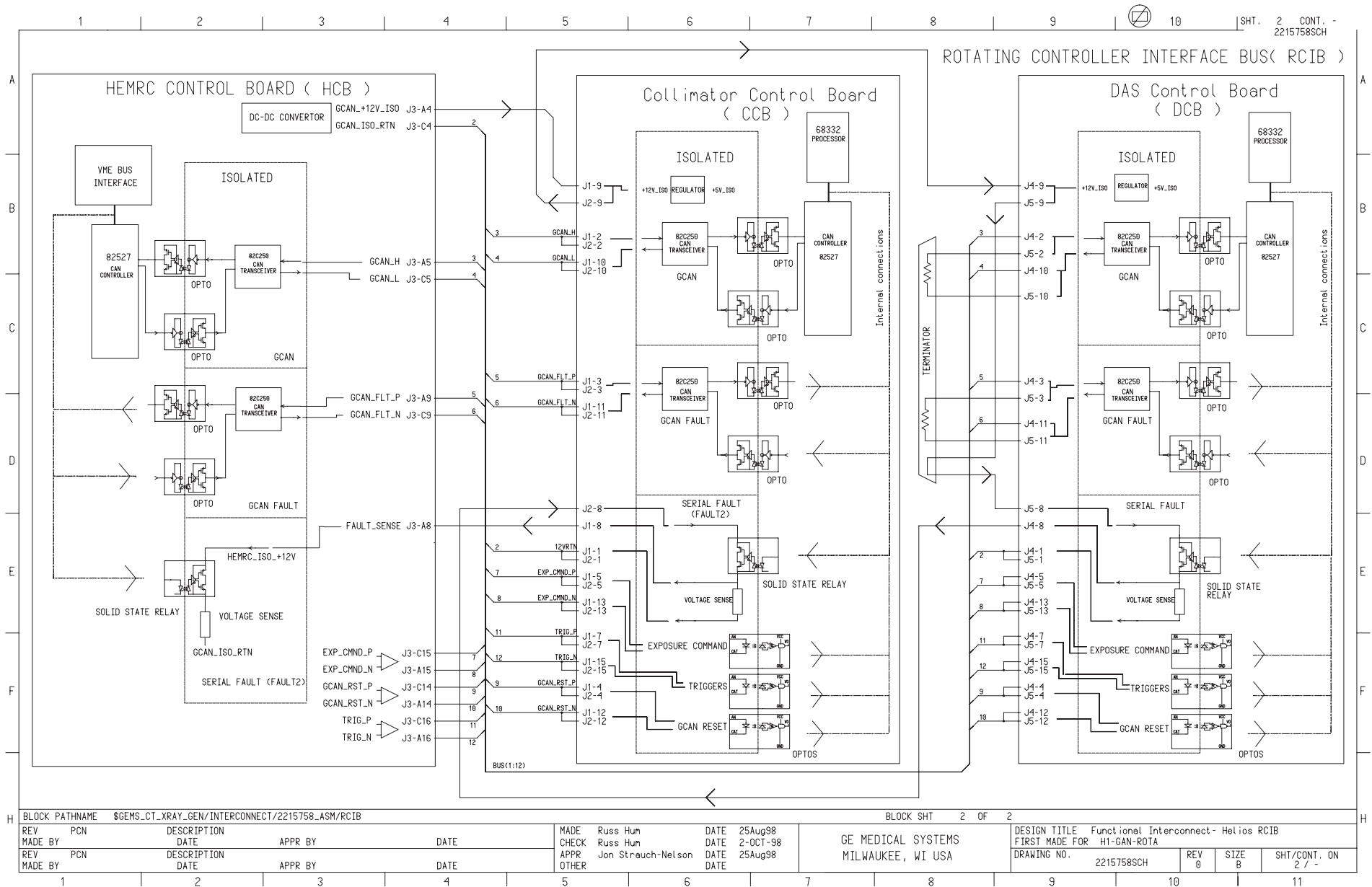


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Section 4.0 - SCU Boot Link / Image Network - 2151851

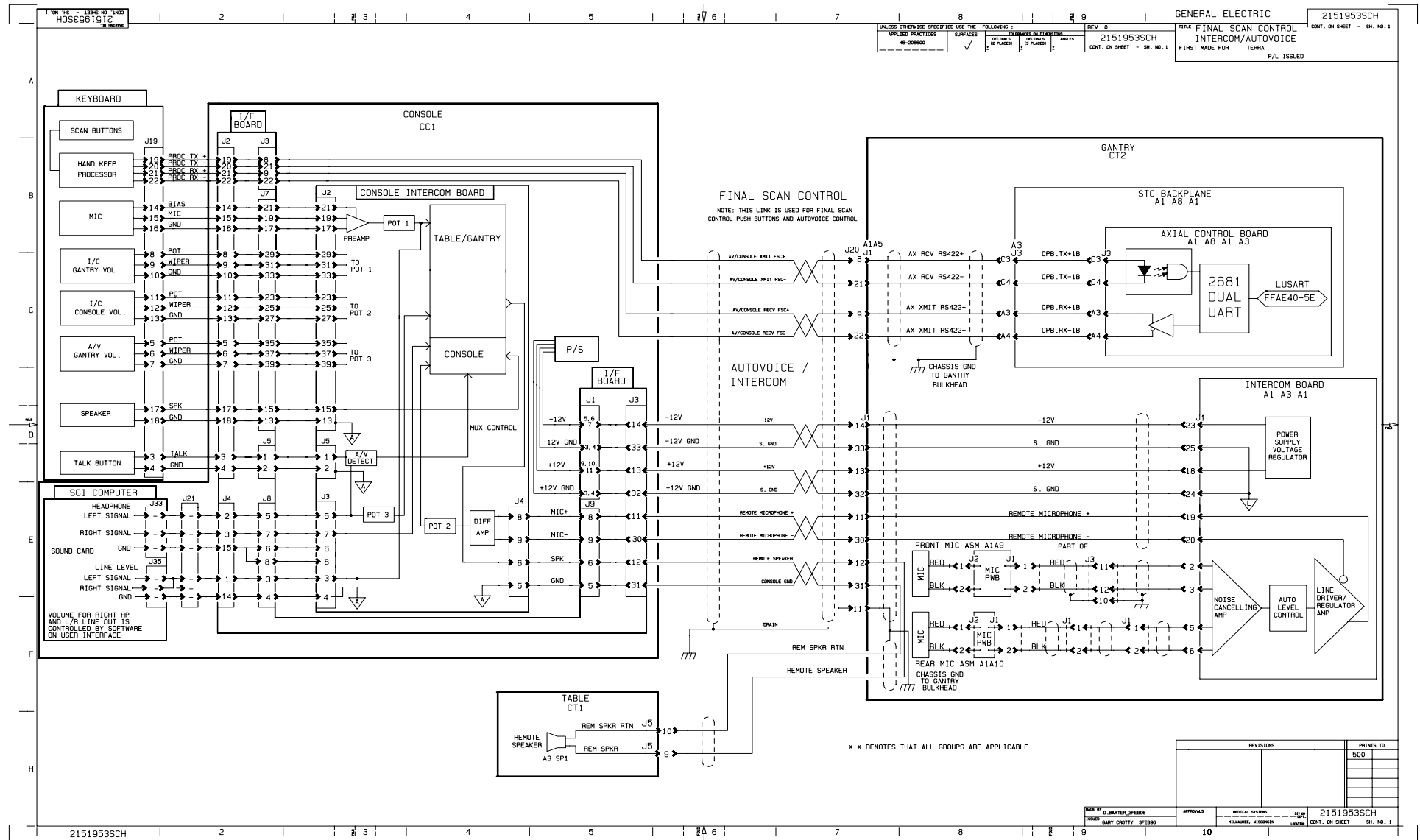


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Section 5.0 - Final Scan Control / Intercom / Autovoice - 2151953

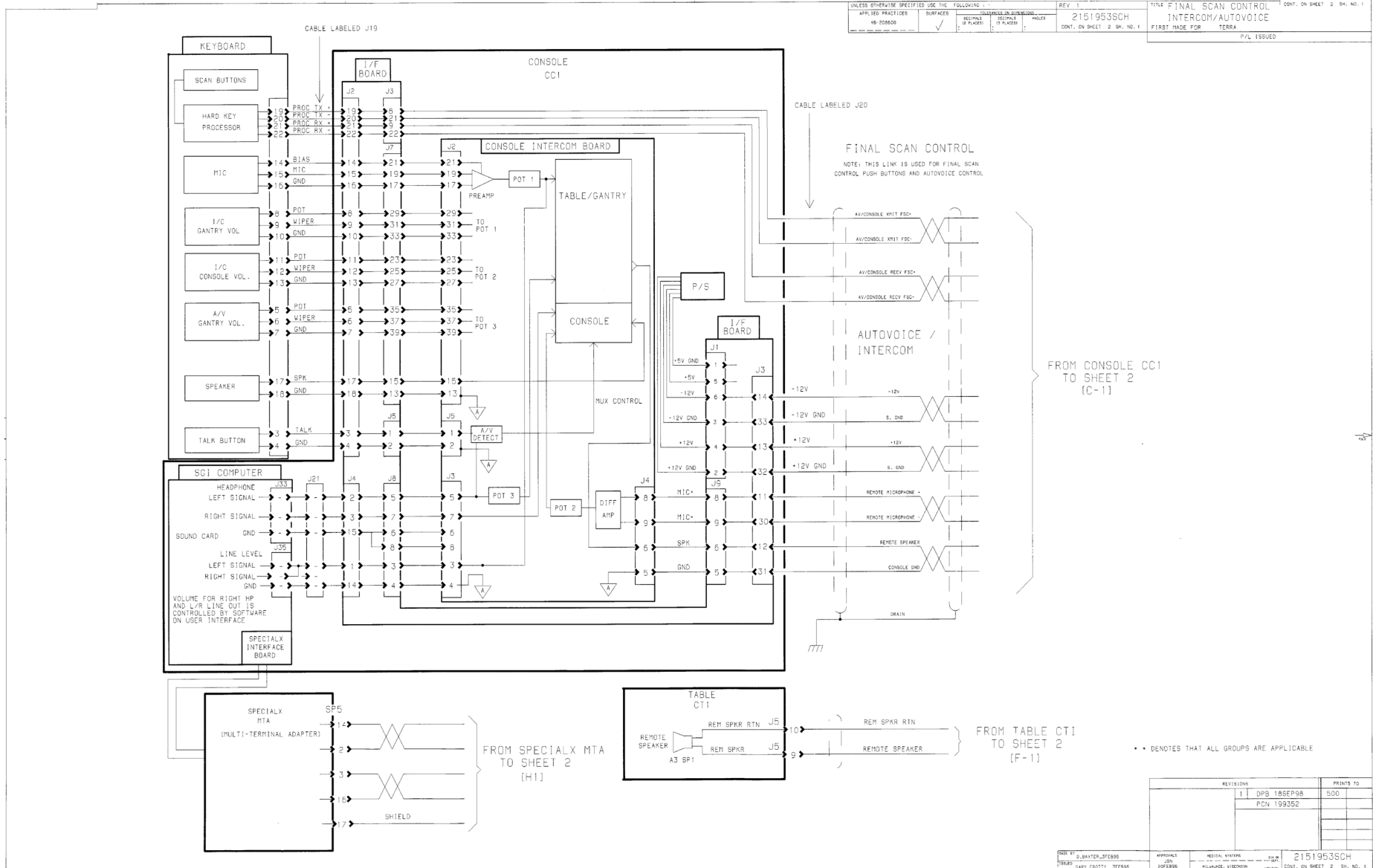


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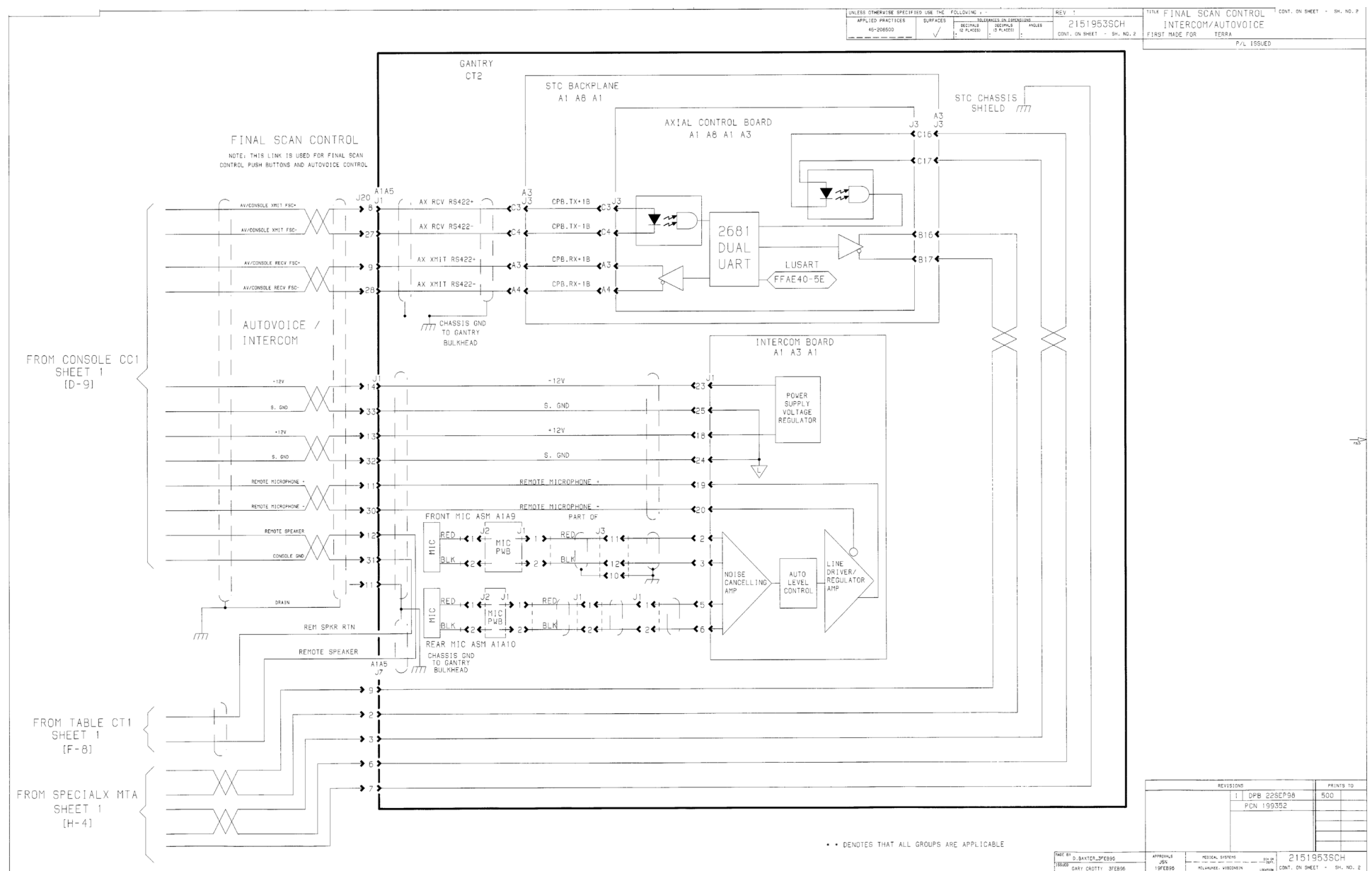
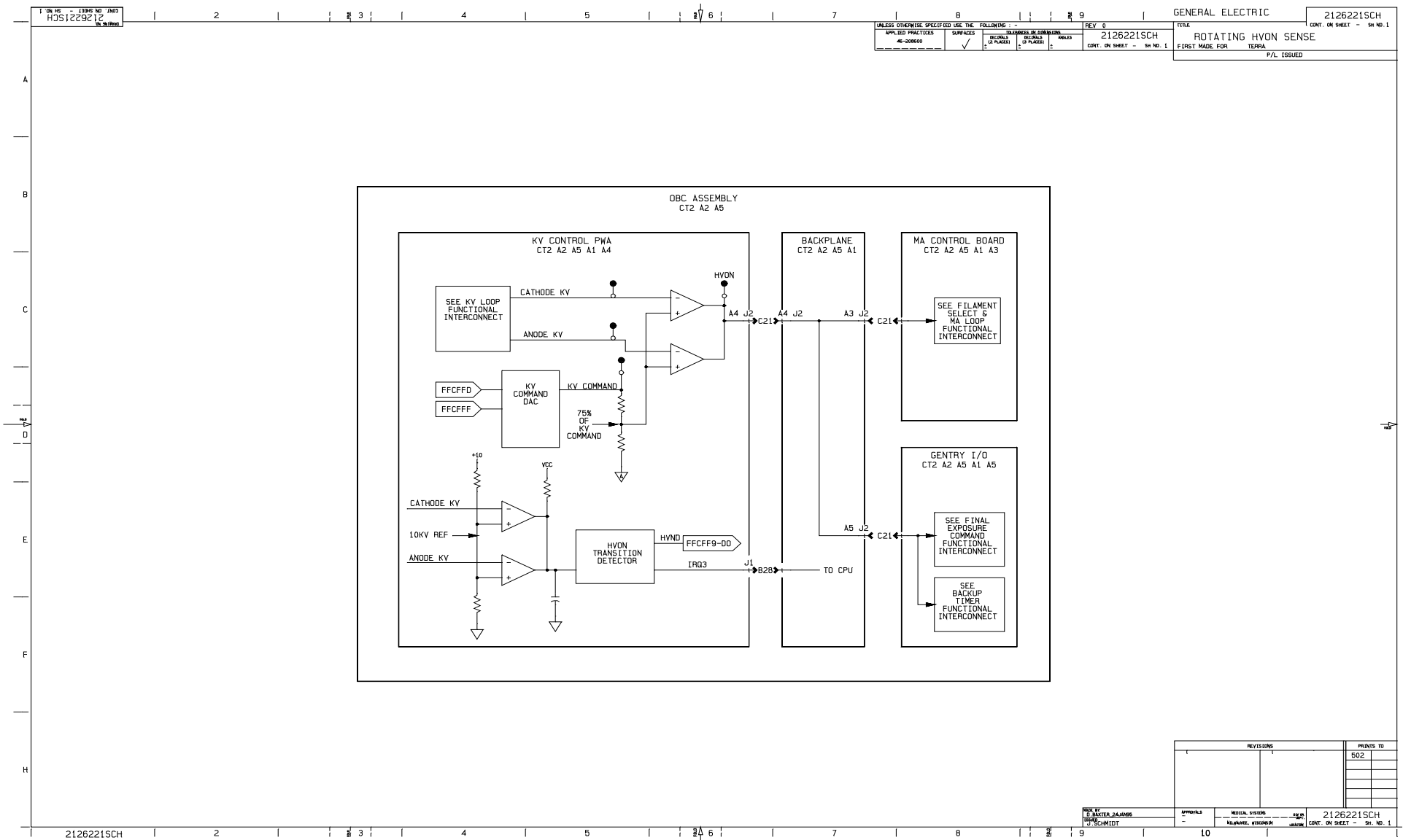


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Chapter 2

X-Ray Generation

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Figure 2-3 2126221 Sheet 1 Rev. 0

Section 4.0 - Filament Select and mA Loop - 2126212

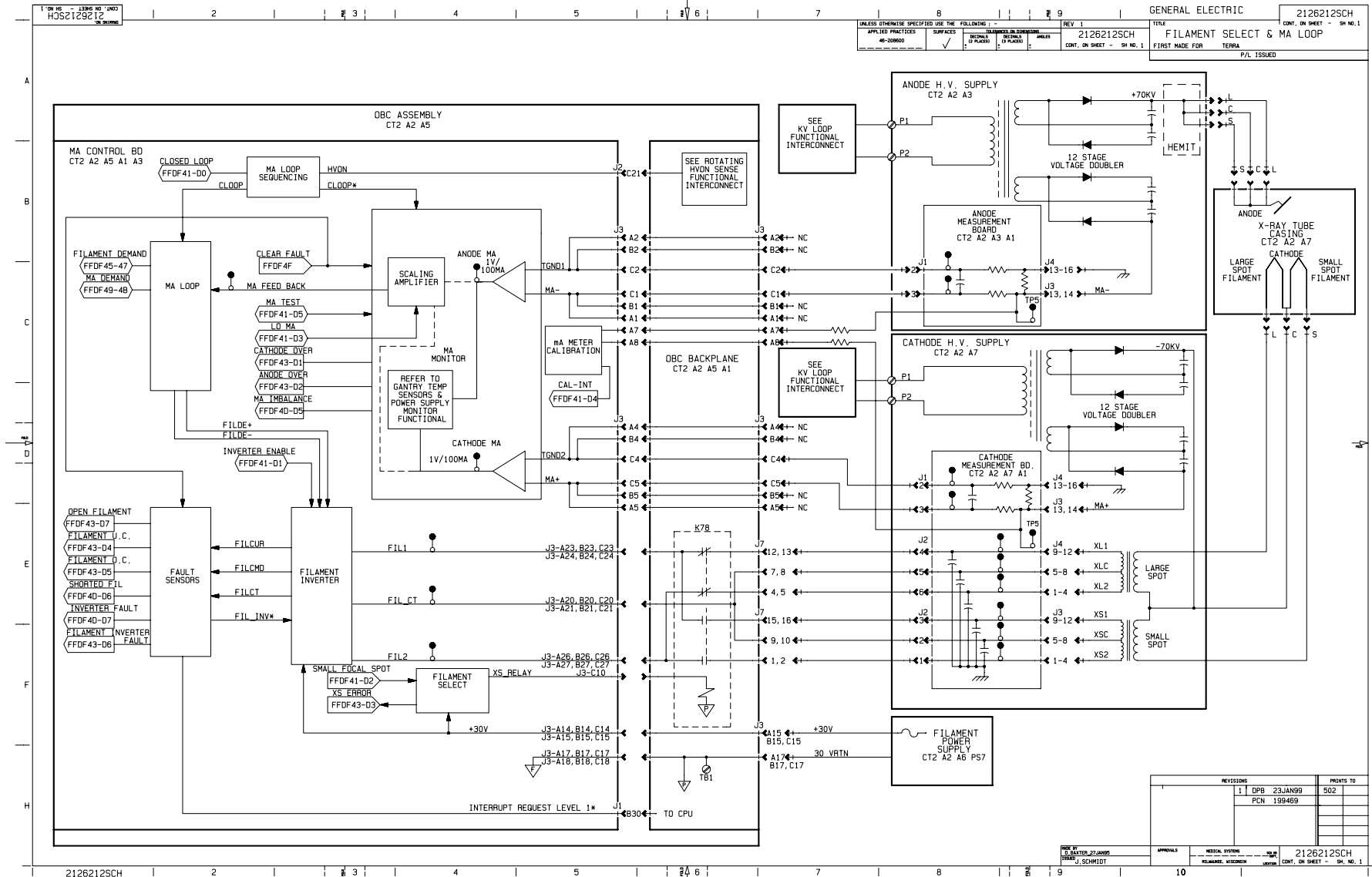


Figure 2-4 2126212 Sheet 1 Rev. 1

Section 5.0 - Tube ID, Cooling and Pressure Sense - 2126213

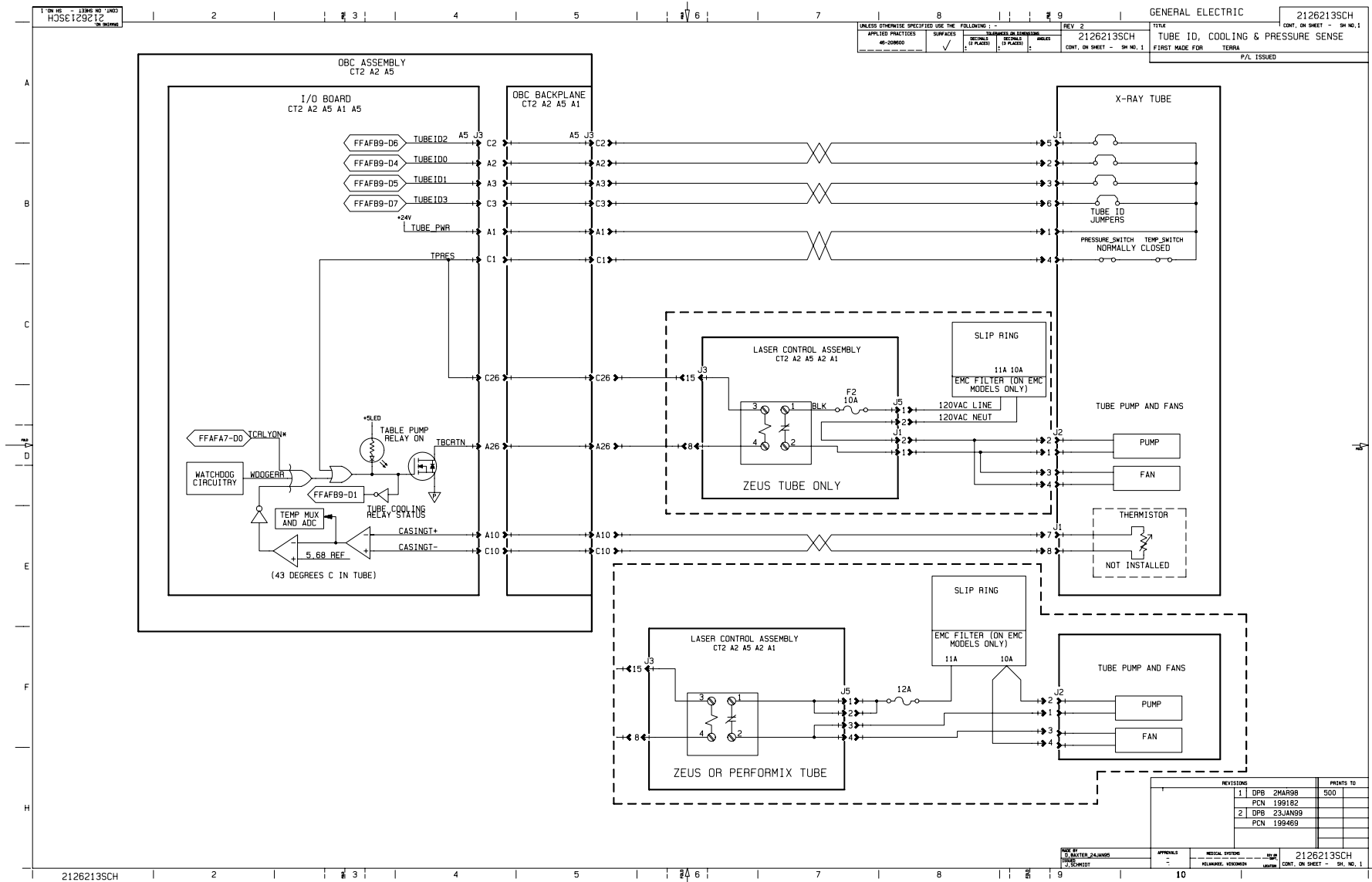
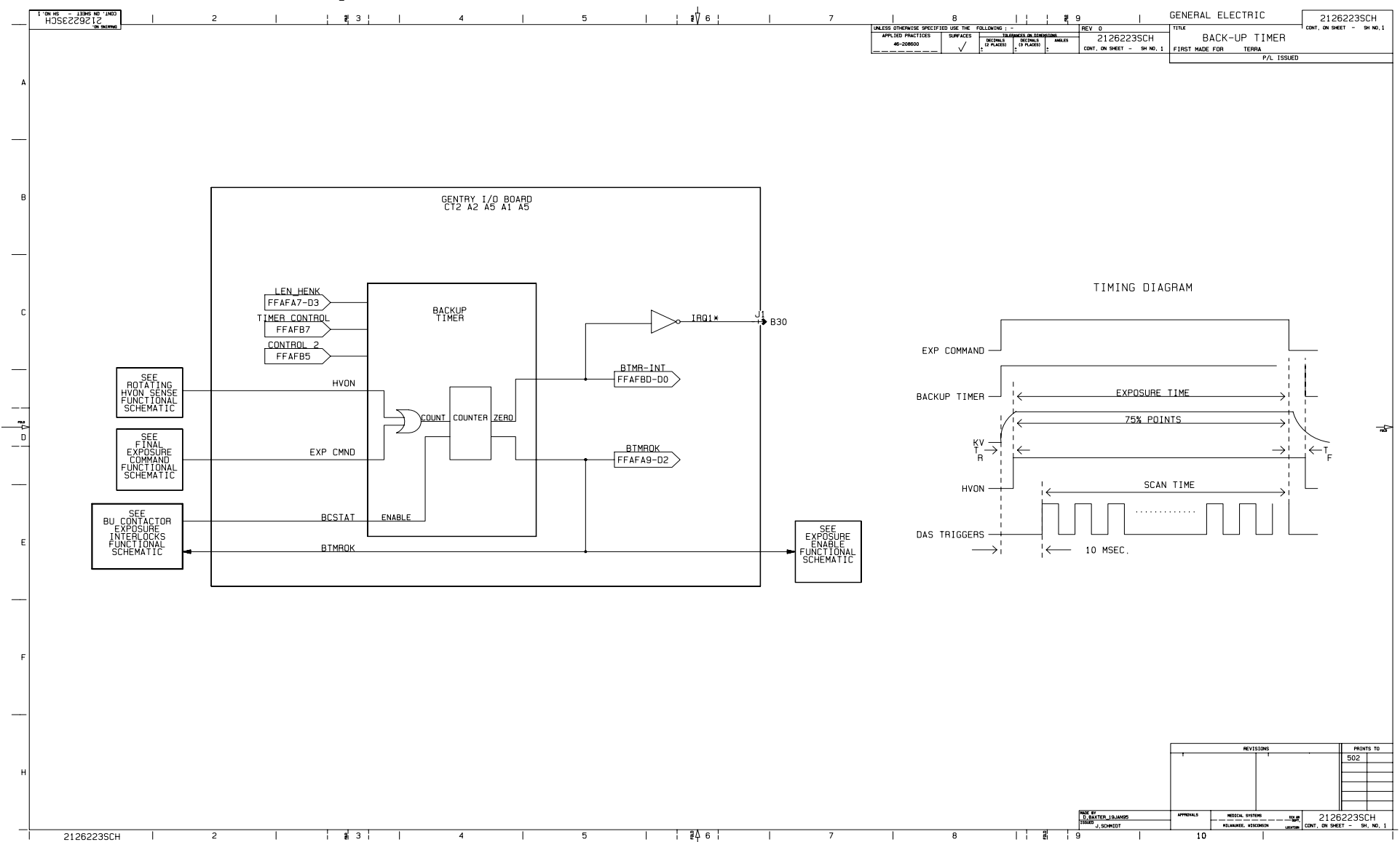


Figure 2-5 2126213 Sheet 1 Rev. 2

Section 6.0 - Back-Up Timer - 2126223



Section 7.0 - kV Loop - 2126211

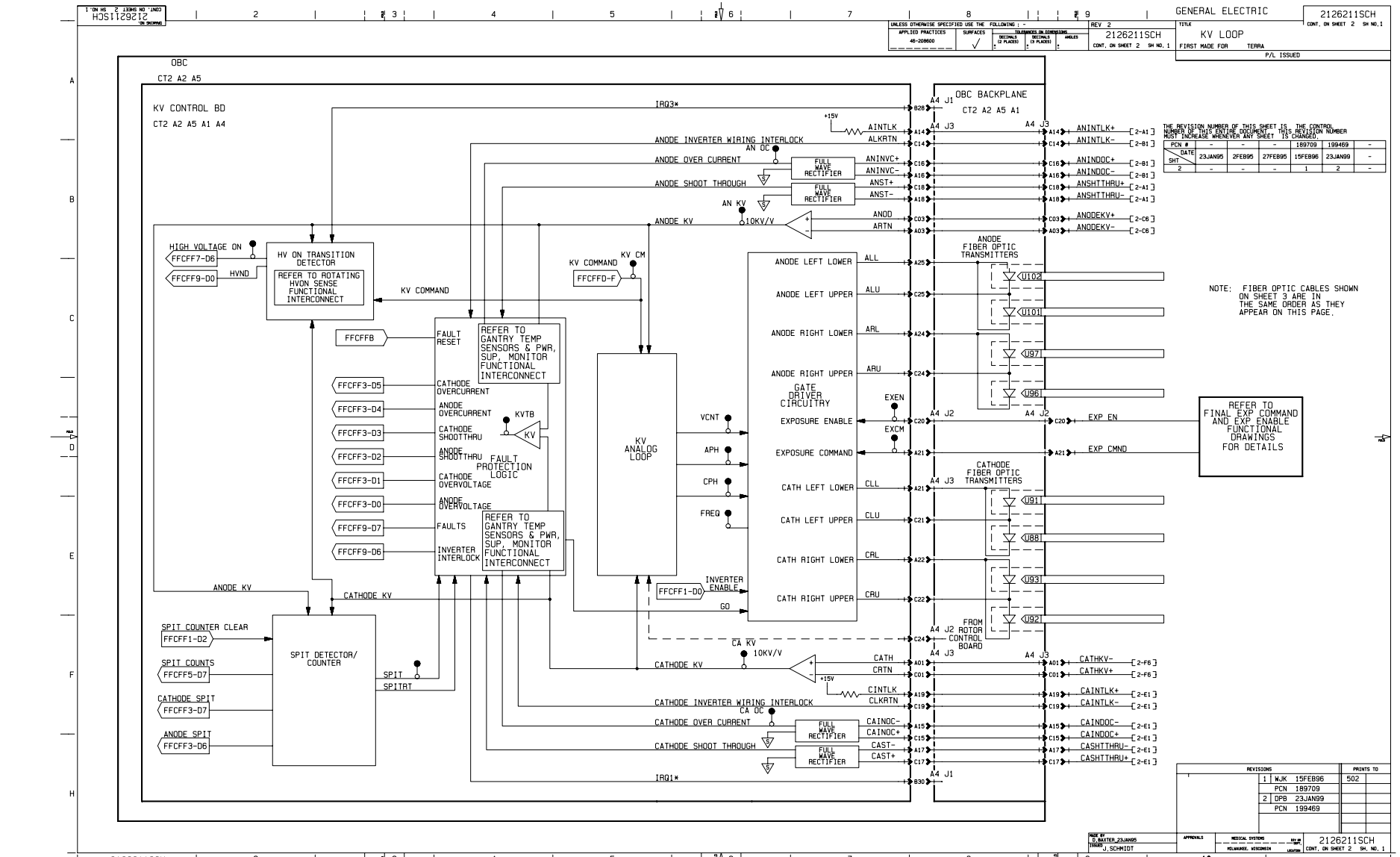


Figure 2-7 2126211 Sheet 1 Rev. 2

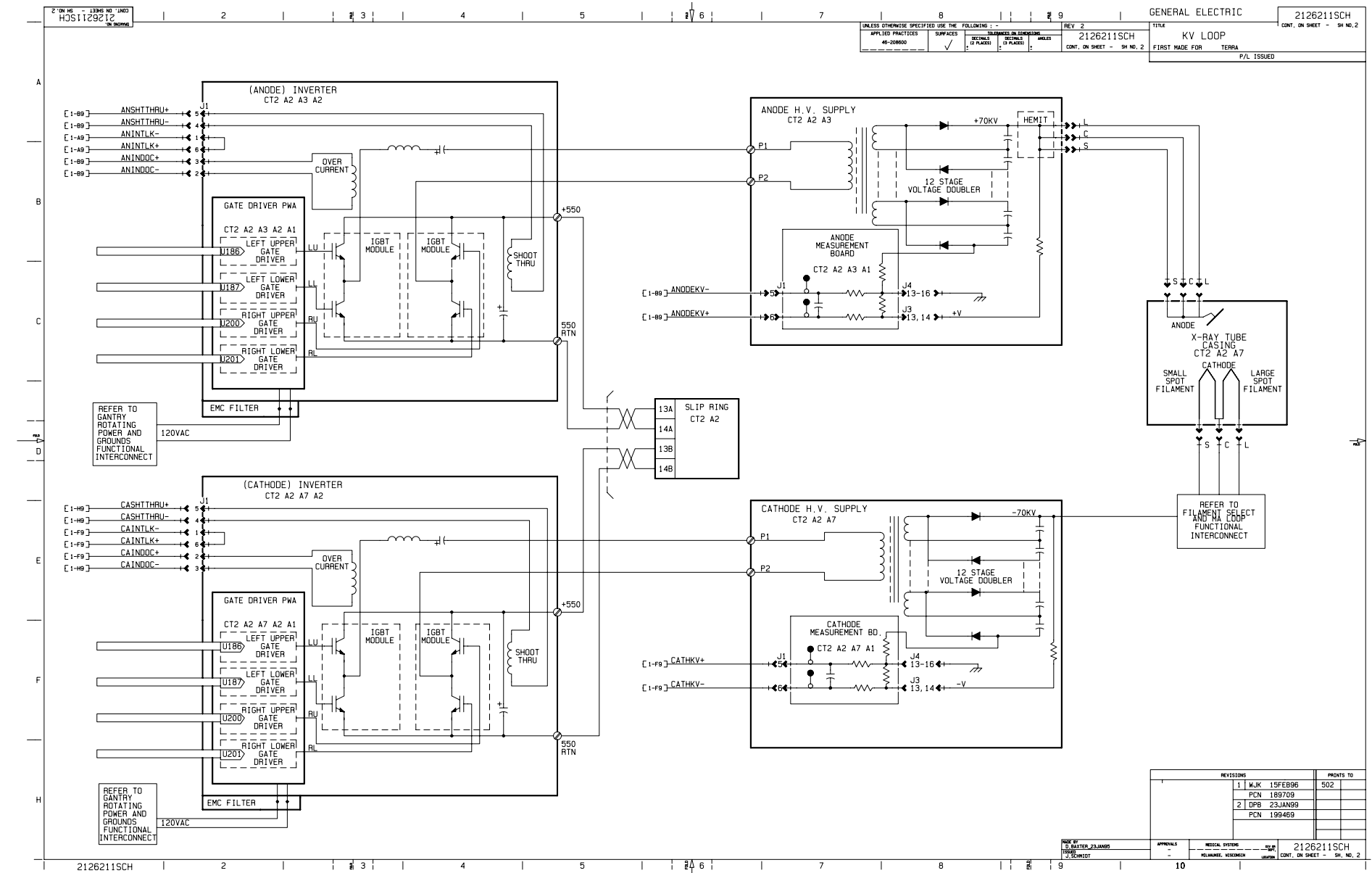


Figure 2-9 2126211 Sheet 3 Rev. 1

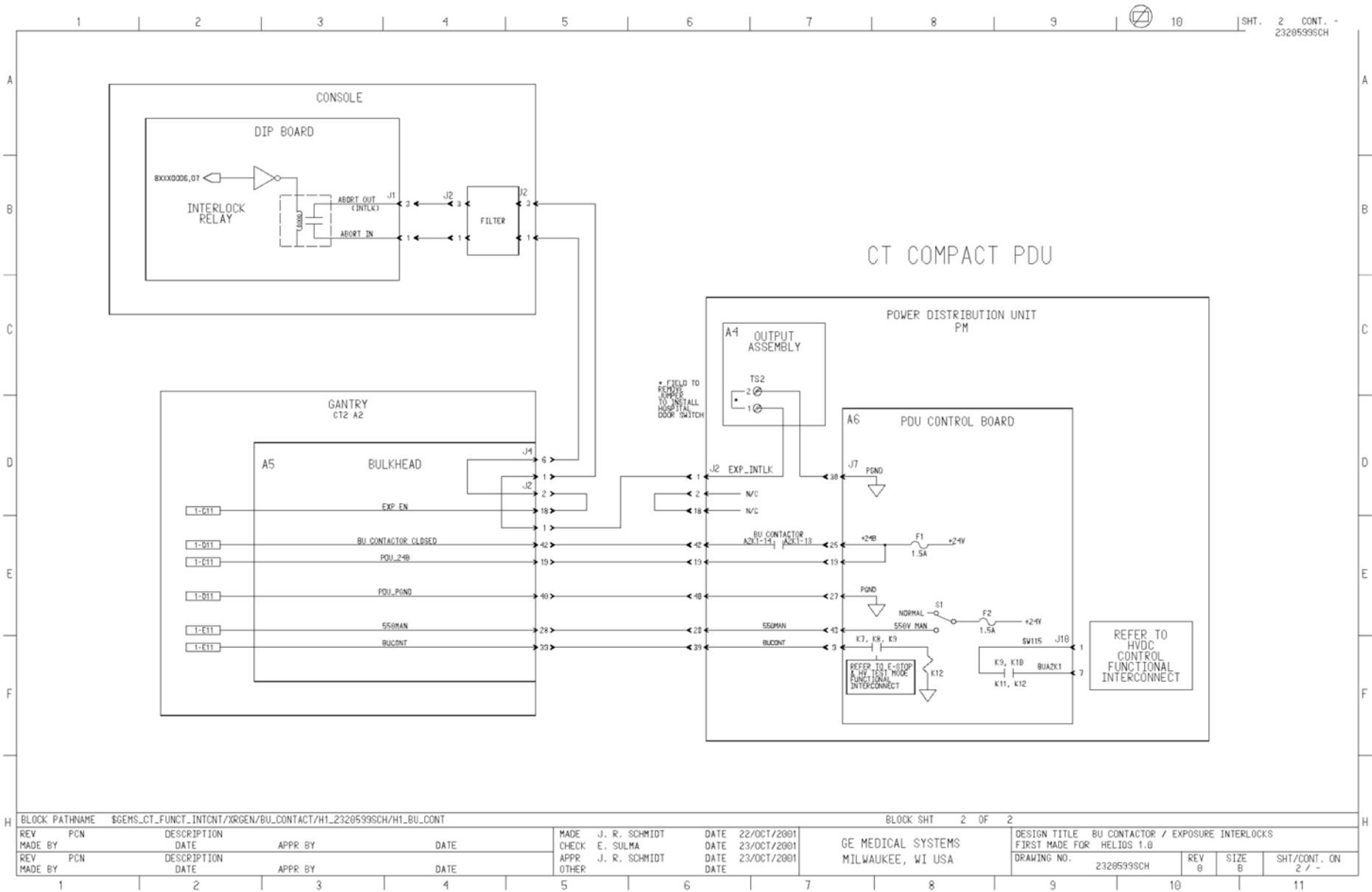


Figure 2-11 2320599 sheet 02, Rev. 0

Section 8.0 - BU Contactor / Exposure Interlocks - 2320599

Page 44

Section 9.0 - Rotor Control - 2216236

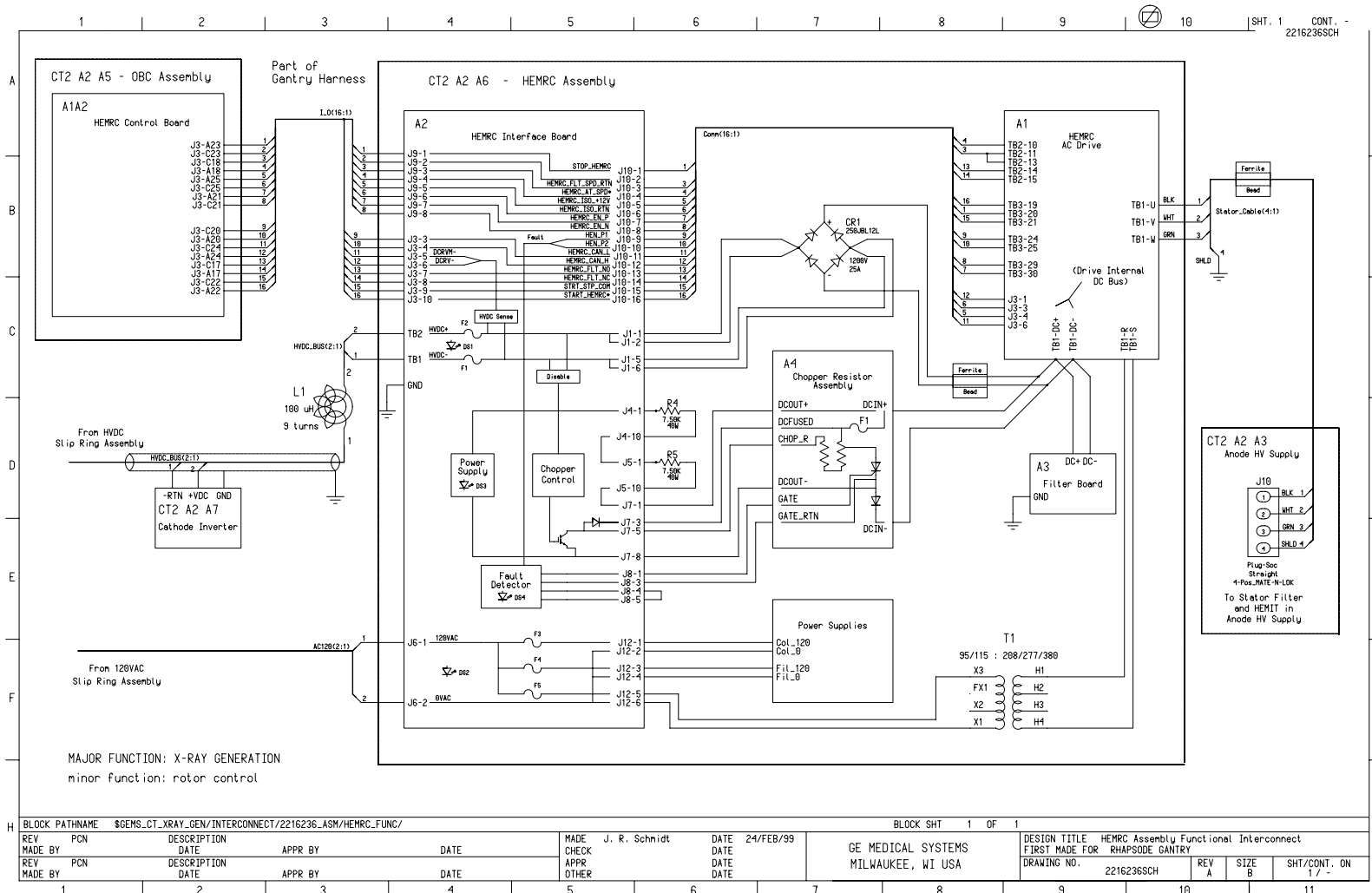


Figure 2-12 2216236 Sheet 1 Rev. 0

Section 10.0 - X-Ray Collimation and Filtration RCIB - 2215631

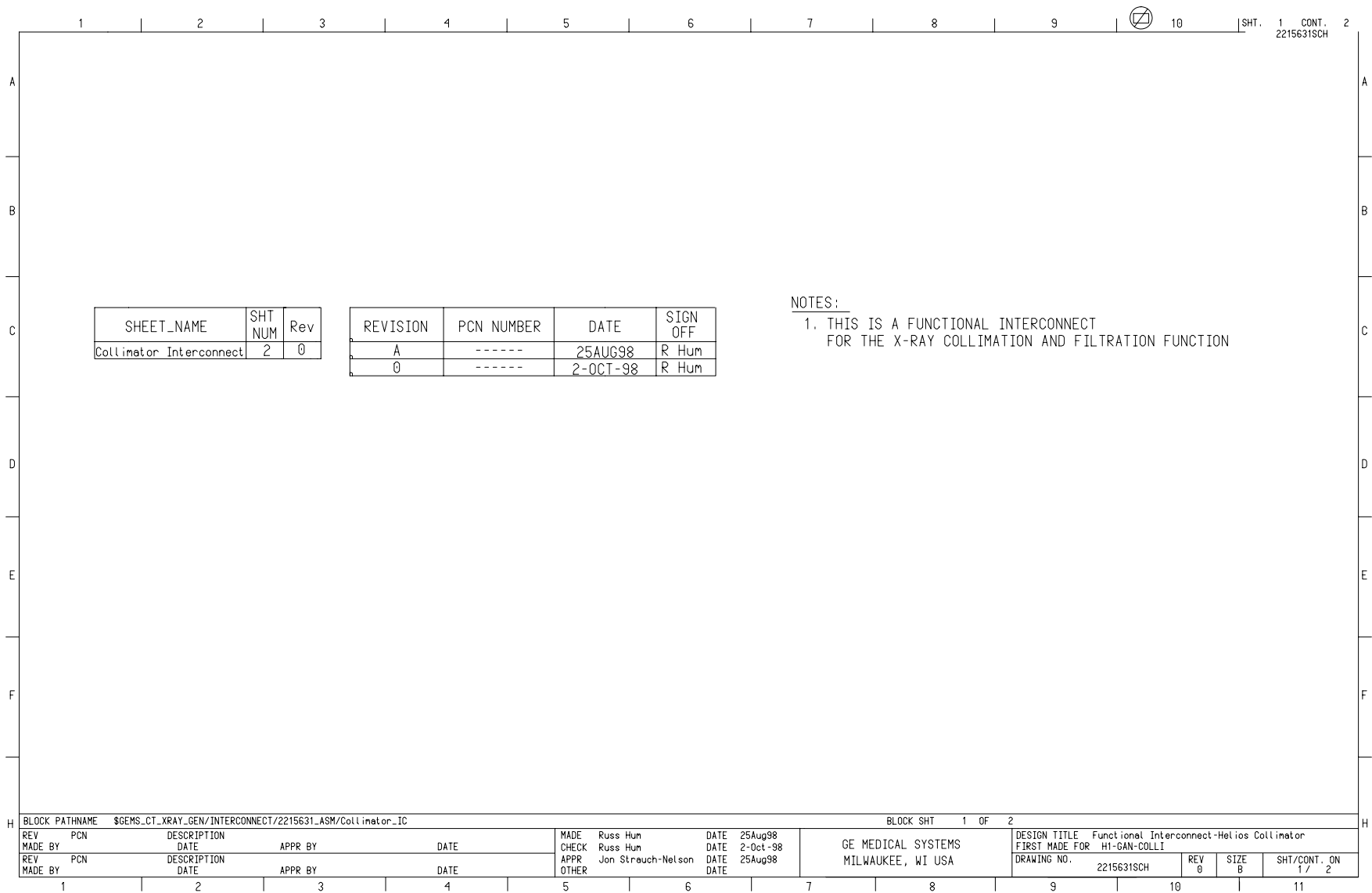
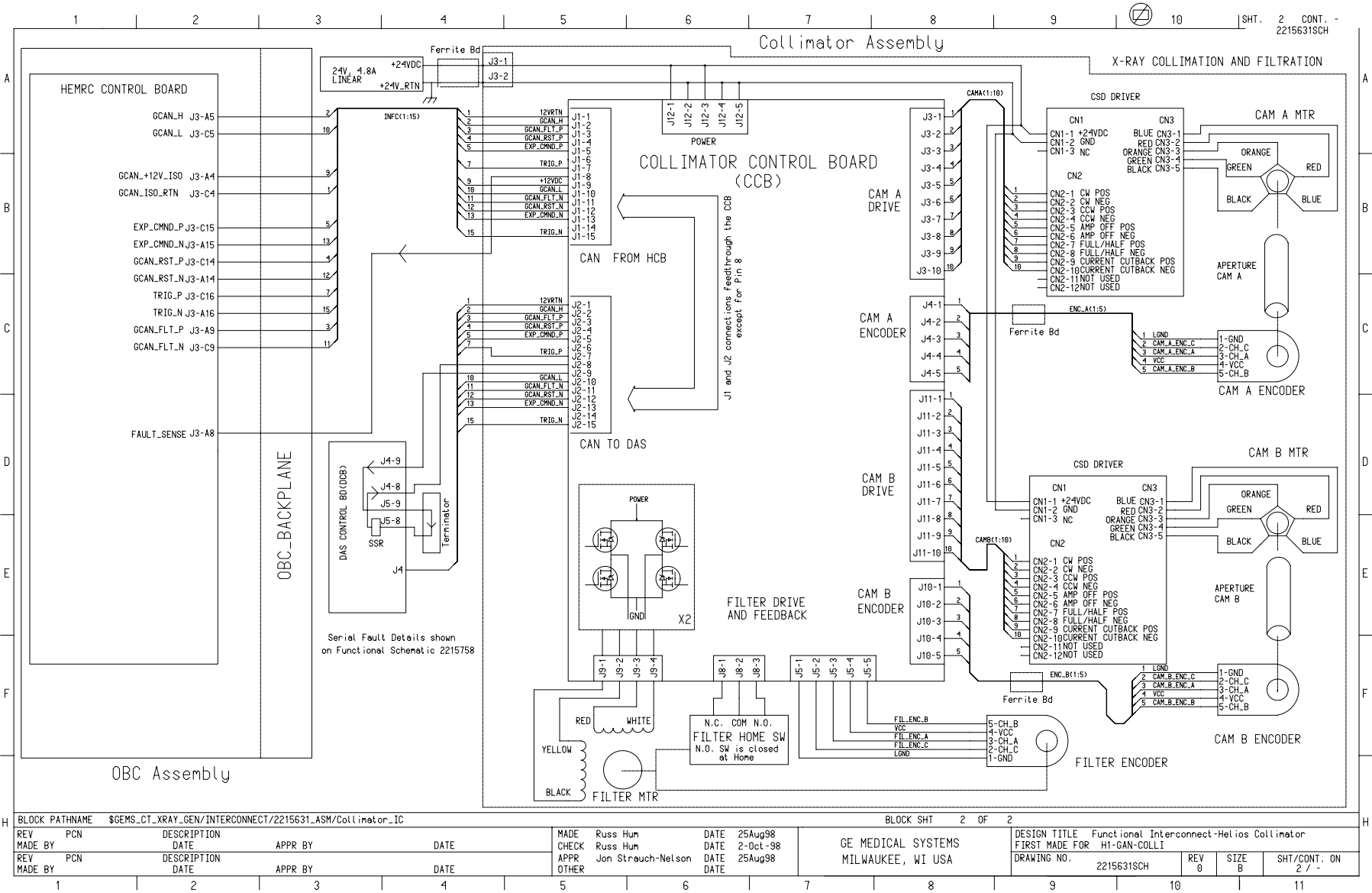


Figure 2-13 2215631 Sheet 1 Rev. 0



Section 11.0 - Table Gantry Sync - 2126191

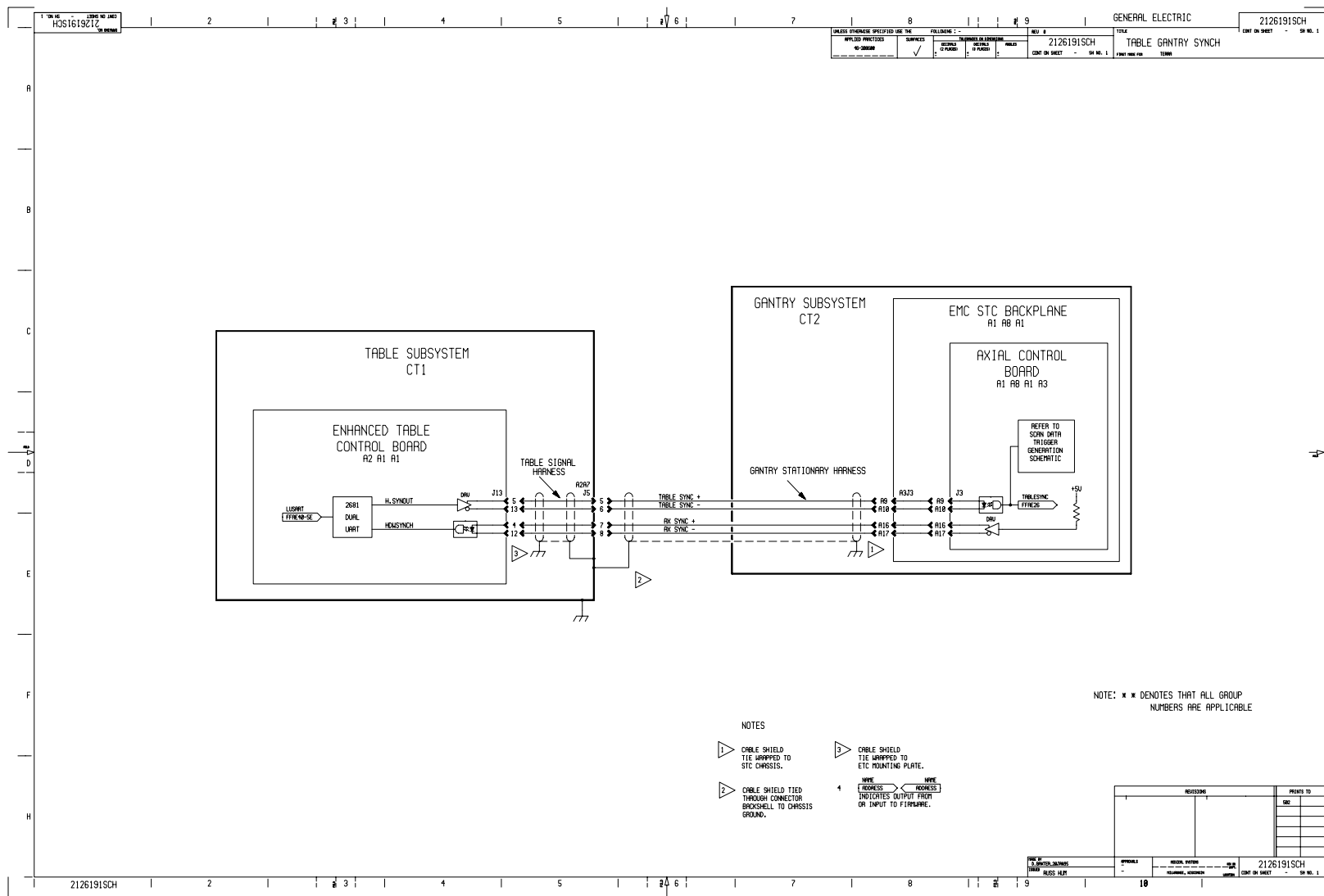


Figure 2-15 2126191 Sheet 1 Rev. 0

Chapter 3

Data Acquisition

Figure 3-1 2126245 Sheet 1 Rev. 1



Section 2.0 - Scan Data Acquisition-FET Set-Up (ETC) - 2217020

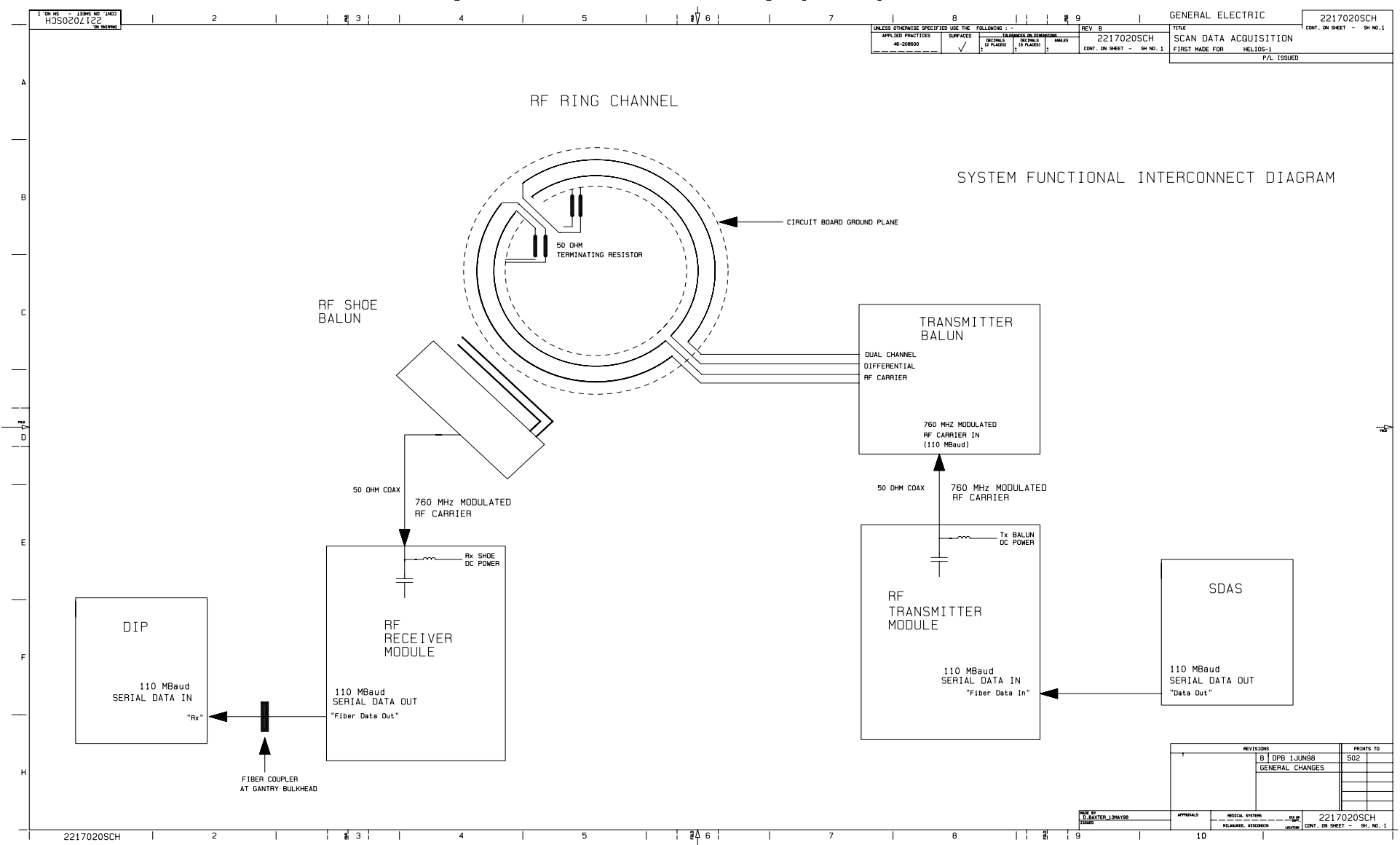


Figure 3-2 2217020 Sheet 1 Rev. B

Section 3.0 - Scan Data Acquisition-FET Set-Up (S/A) - 2269964

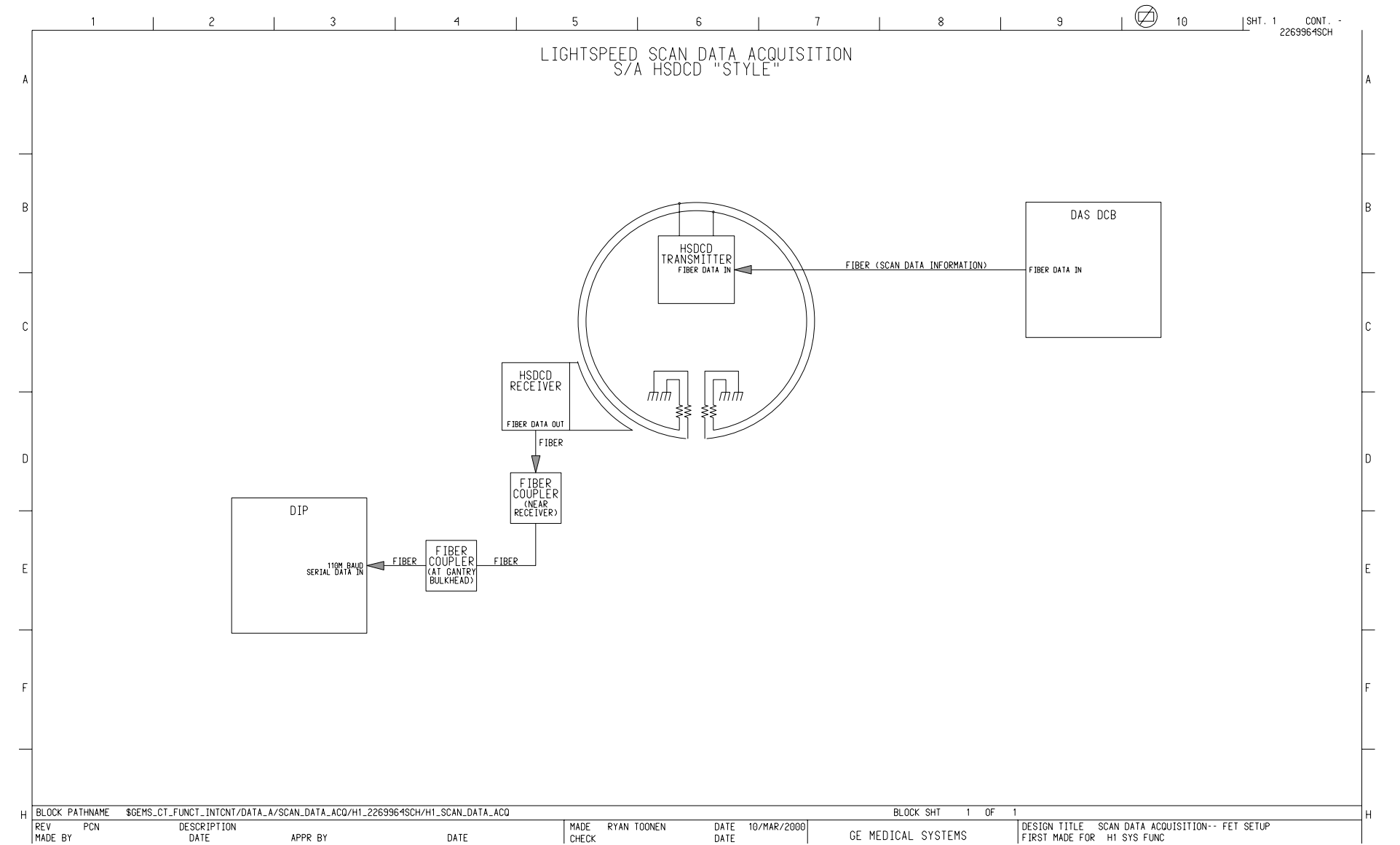


Figure 3-3 2269964 Sheet 1 Rev. A



Chapter 4

Patient Positioning

Section 2.0 - Table Elevation and Tilt Drive - 2126210

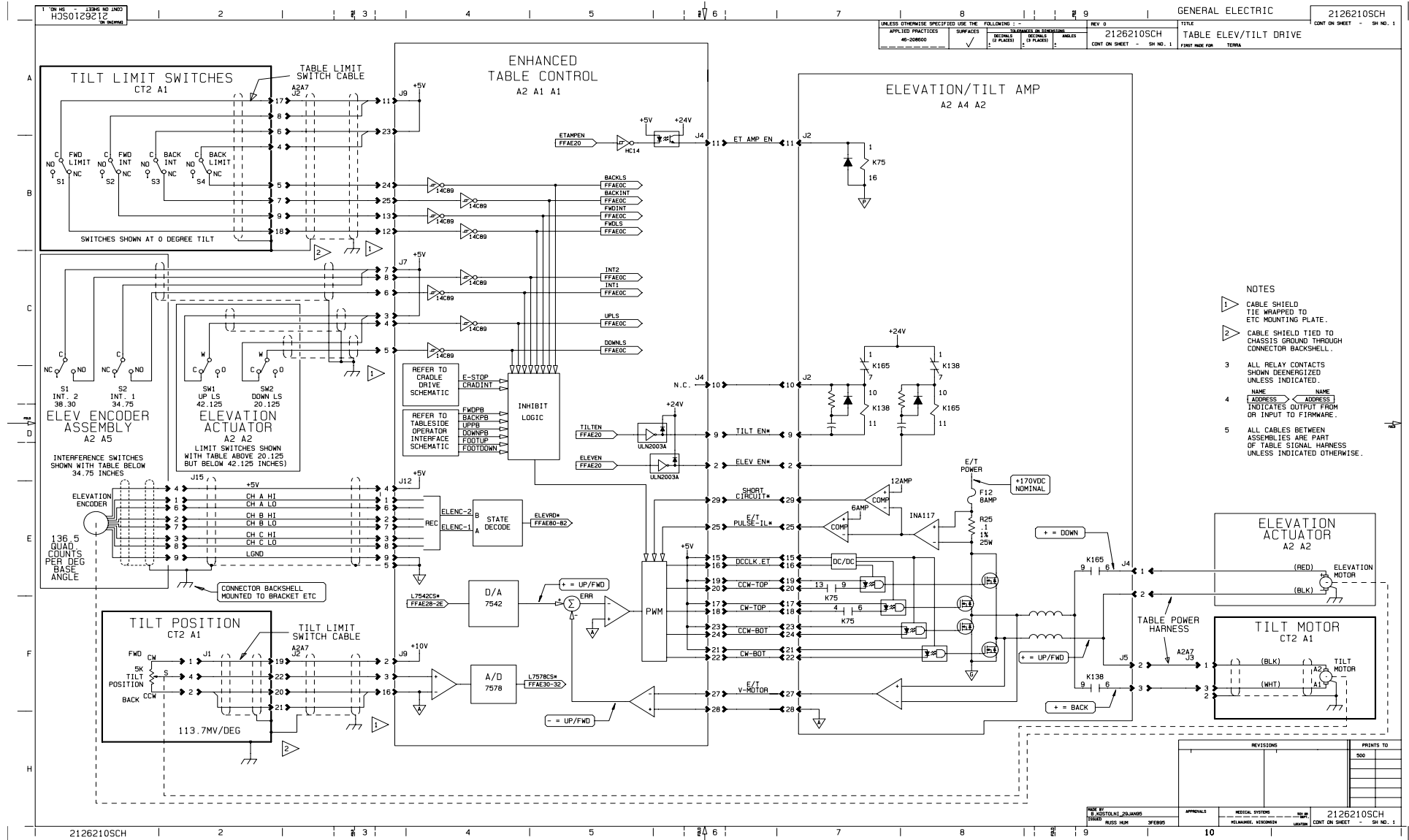


Figure 4-2 2126210 Sheet 1 Rev. 0

Section 3.0 - Table Cradle Drive (Vertical & Longitudinal Drives) - 2123540

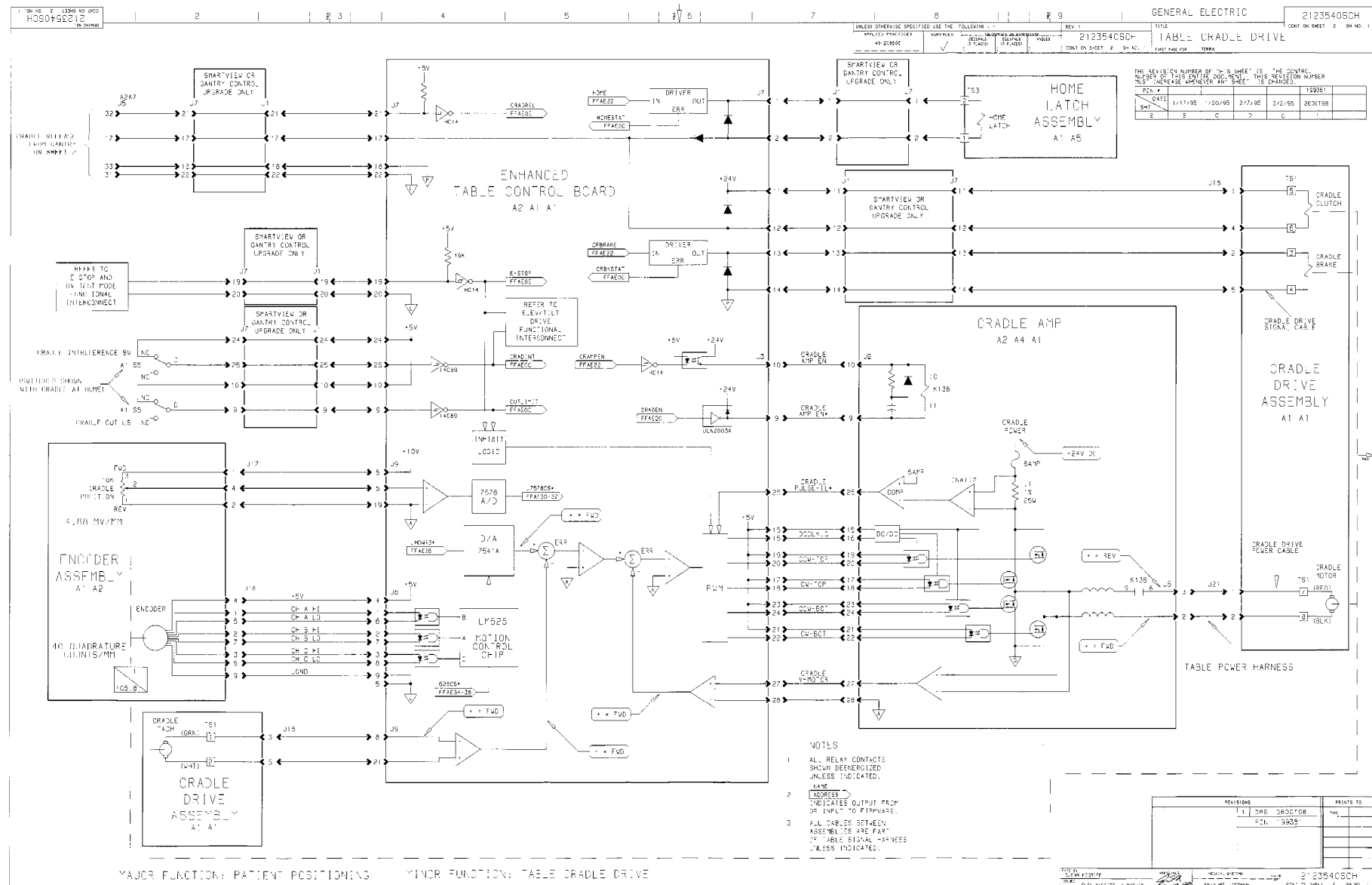


Figure 4-3 2123540 Sheet 1 Rev 1

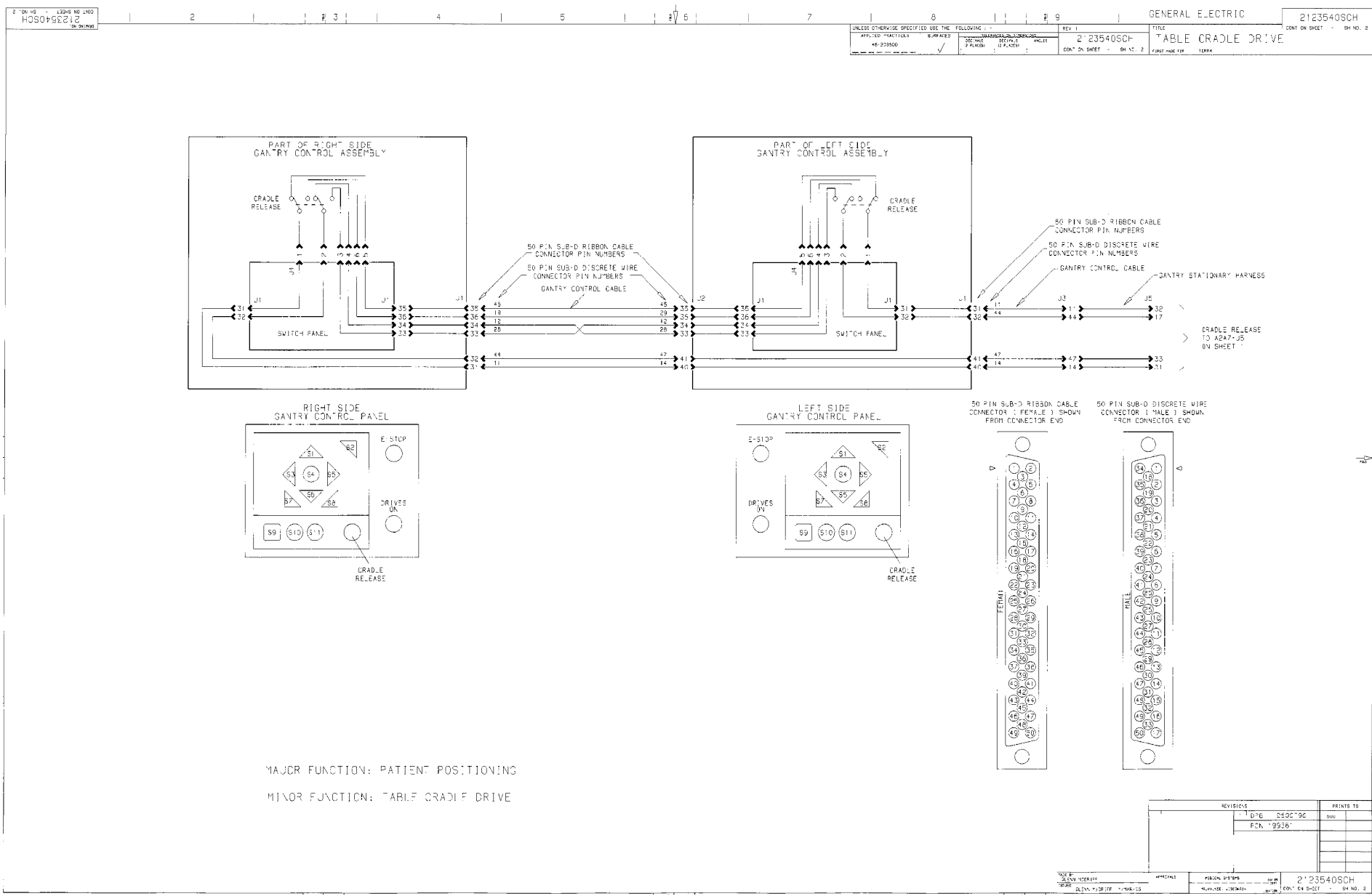


Figure 4-4 2123540 Sheet 2 Rev. 1

Chapter 5

Axial Control

Figure 5-3 2126225 Sheet 2 Rev. 2

Figure 5-4 2126225 Sheet 3 Rev. 1

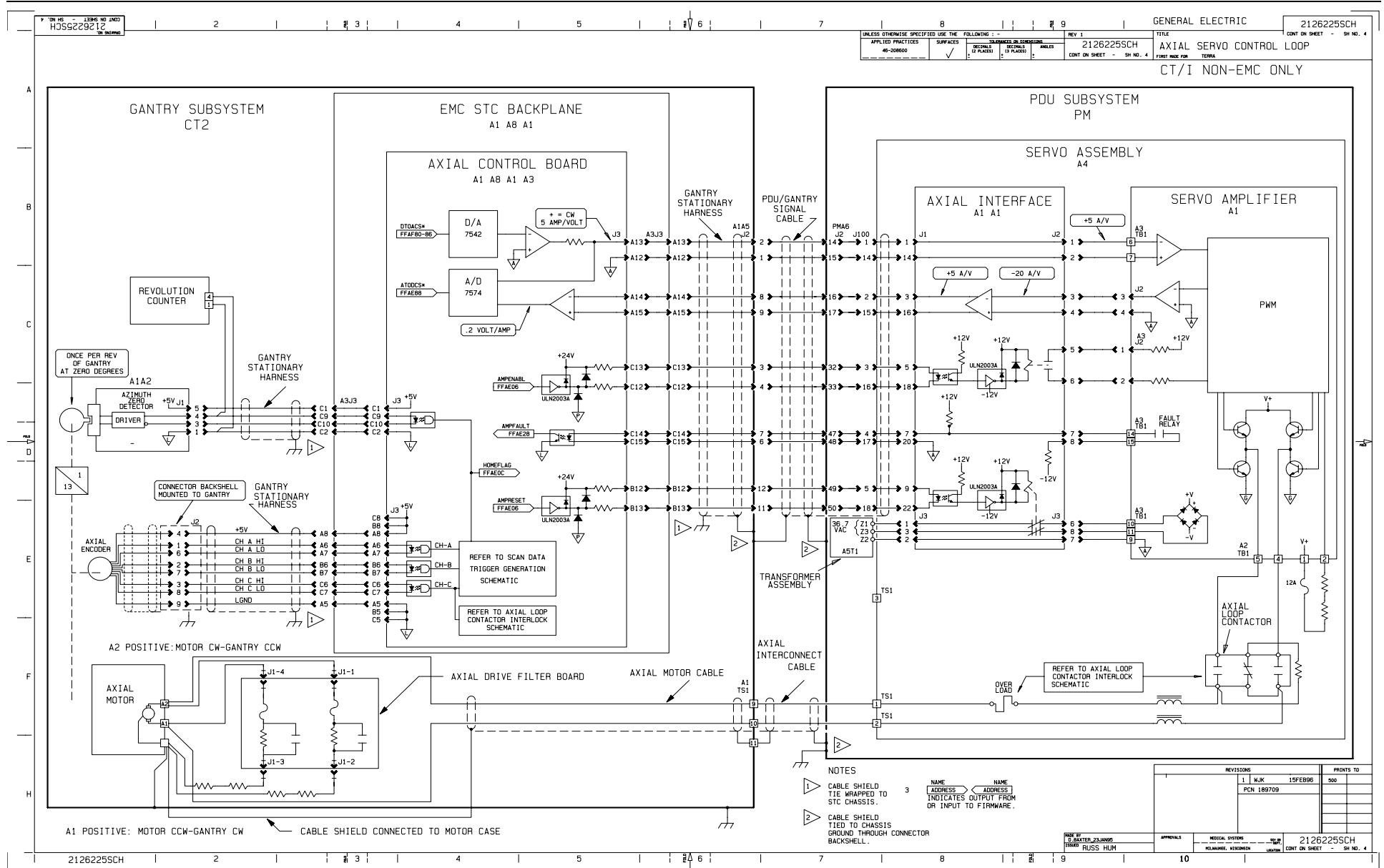


Figure 5-5 2126225 Sheet 4 Rev. 1

Chapter 6

Operator Input/Output

Section 1.0 - Gantry X-Ray Light - 2126182

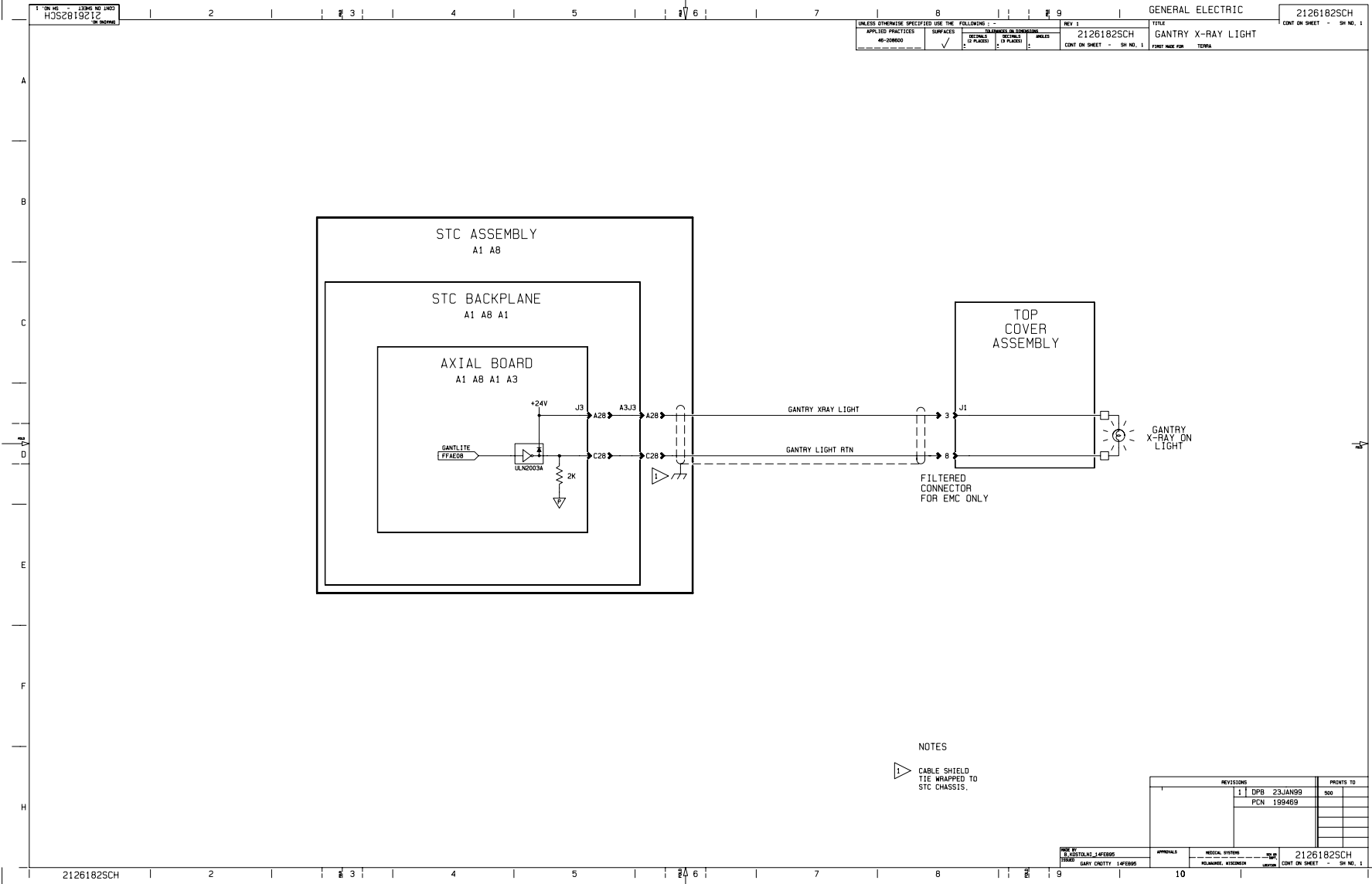


Figure 6-1 2126182 Sheet 1 Rev. 1

Section 3.0 - Table Operator Interface - 2126209

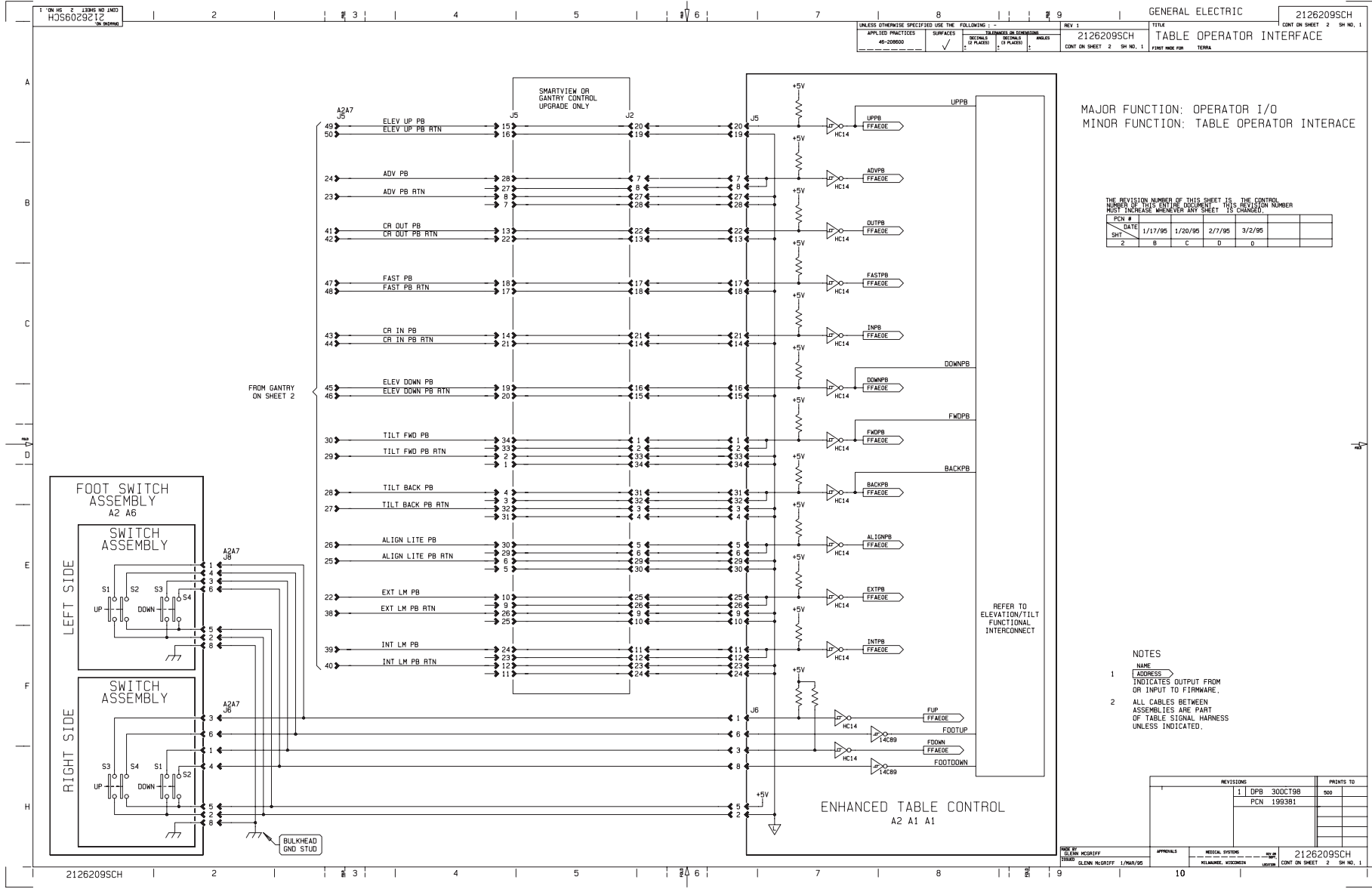


Figure 6-3 2126209 Sheet 1 Rev.1

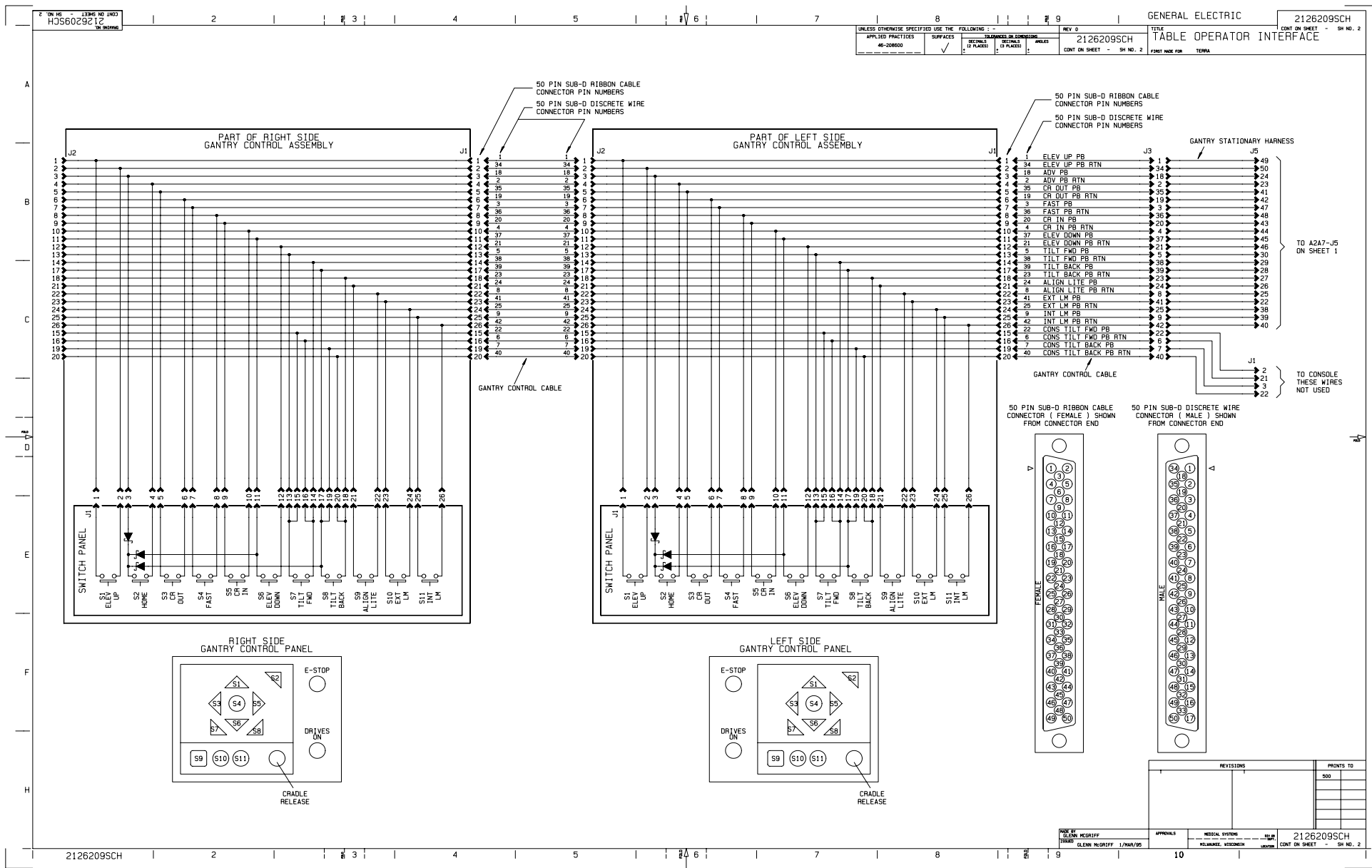


Figure 6-4 2126209 Sheet 2 Rev.0

Chapter 7

System Monitoring

Section 1.0 - Mains Undervoltage - 2126186

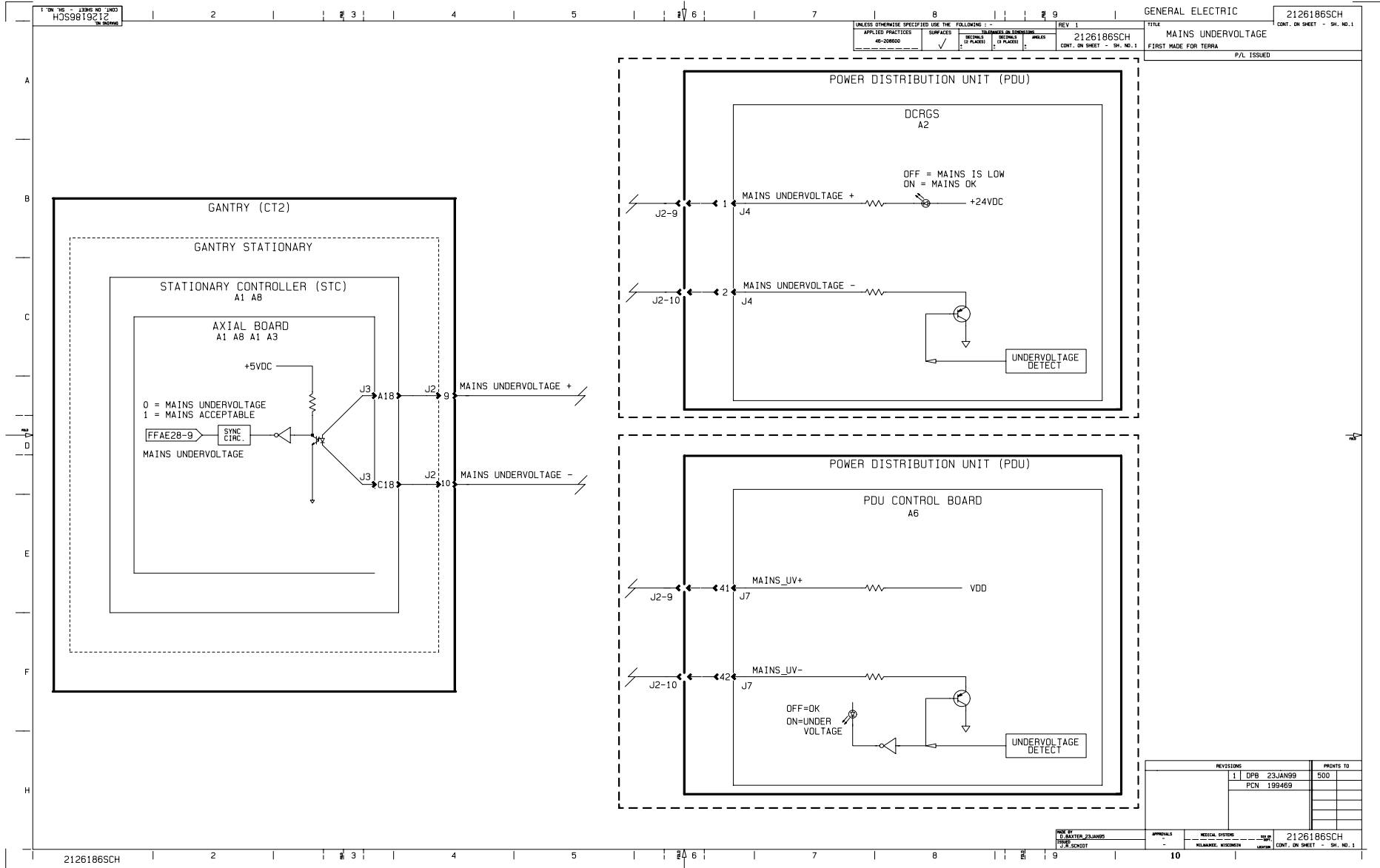


Figure 7-1 2126186 Sheet 1 Rev 1

Section 2.0 - Gantry Temperature Sensors and Power Supplies - 2235161

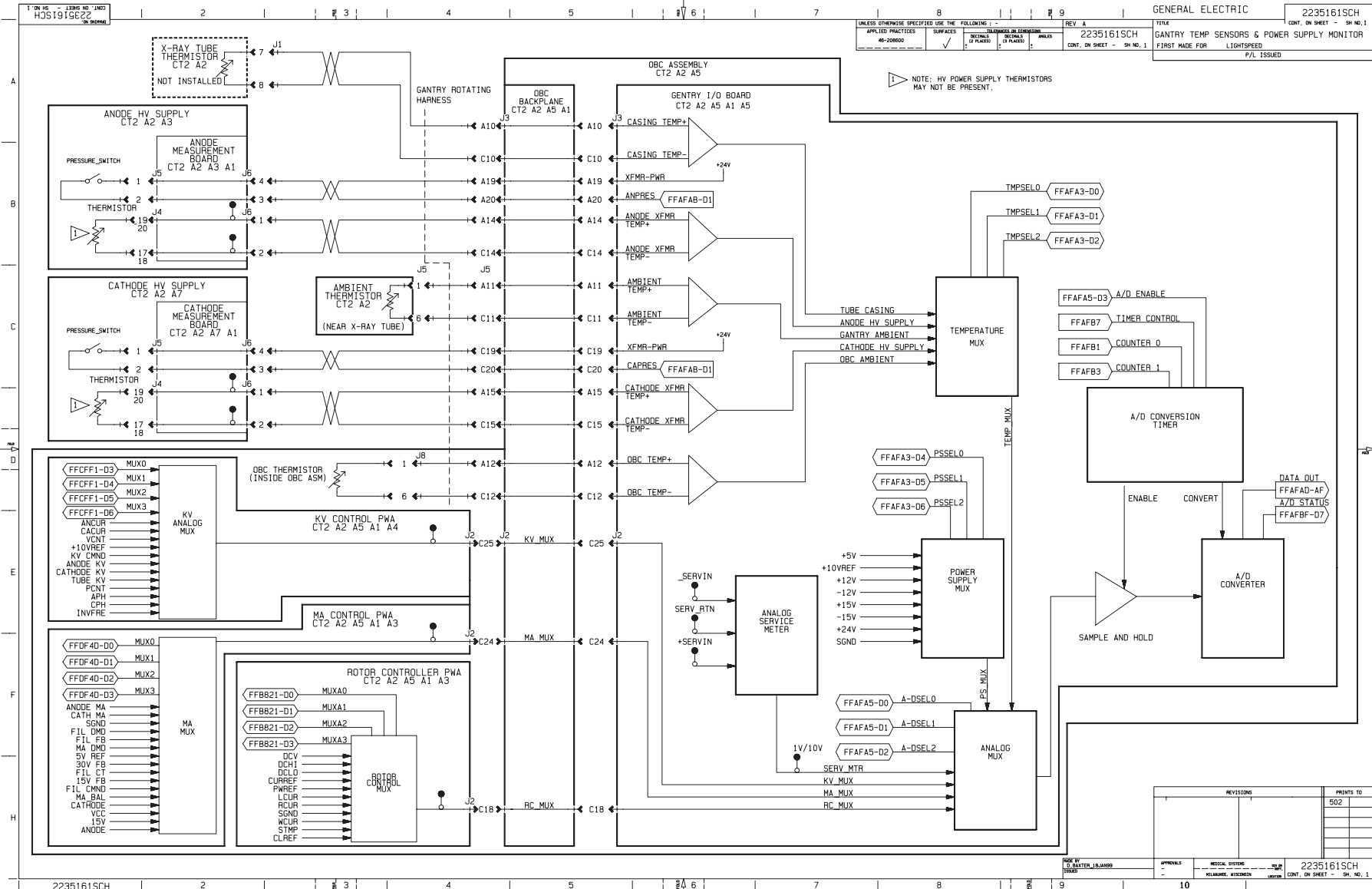


Figure 7-2 2235161 Sheet 1 Rev A

Chapter 8

System Power Control

Section 1.0 - HV Test Mode / E- Stop - 2126206

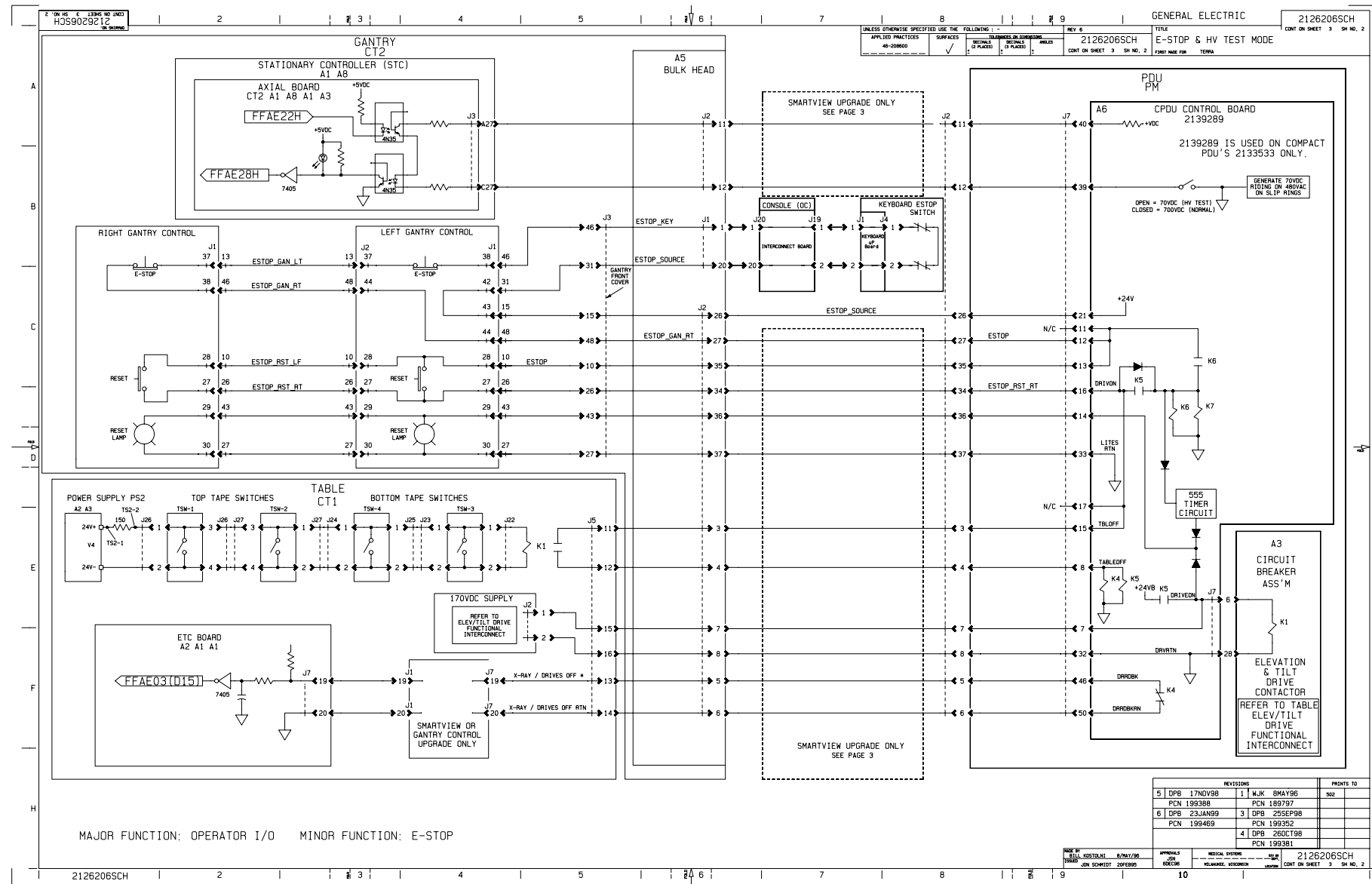


Figure 8-1 2126206 Sheet 2 Rev. 6

Section 2.0 - HVDC Bus Control - 2126183

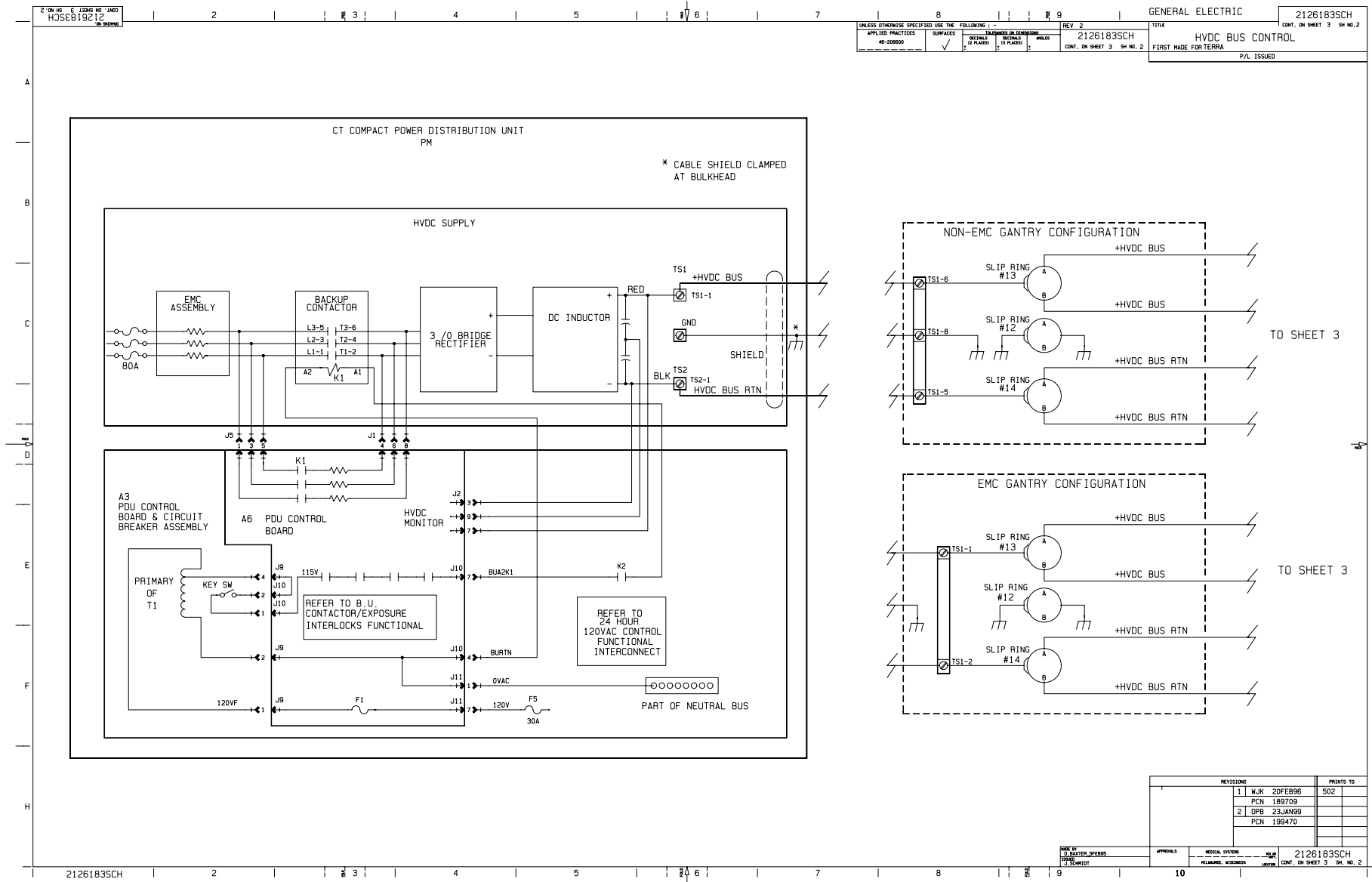


Figure 8-2 2126183 Sheet 2 Rev 2

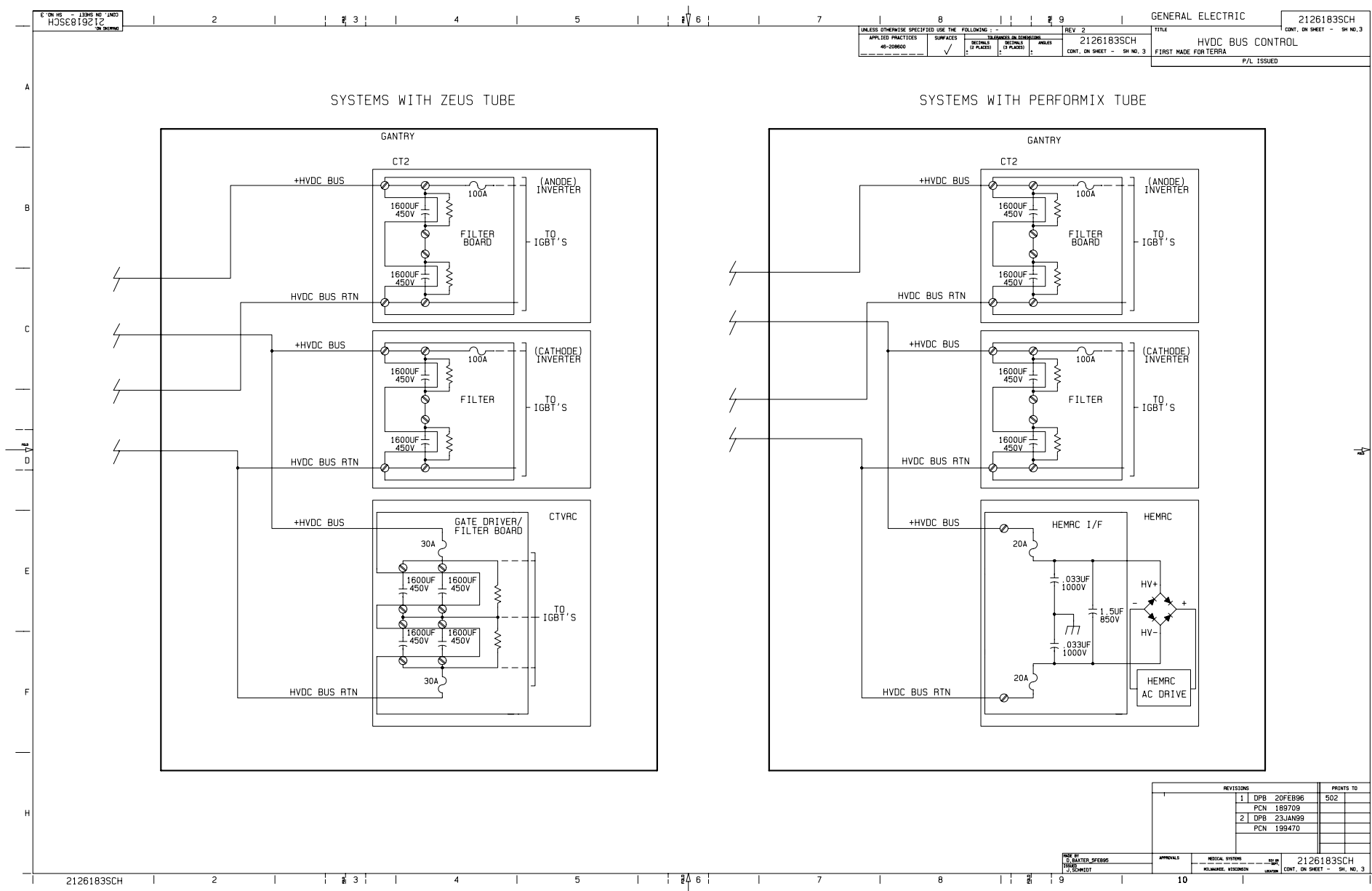
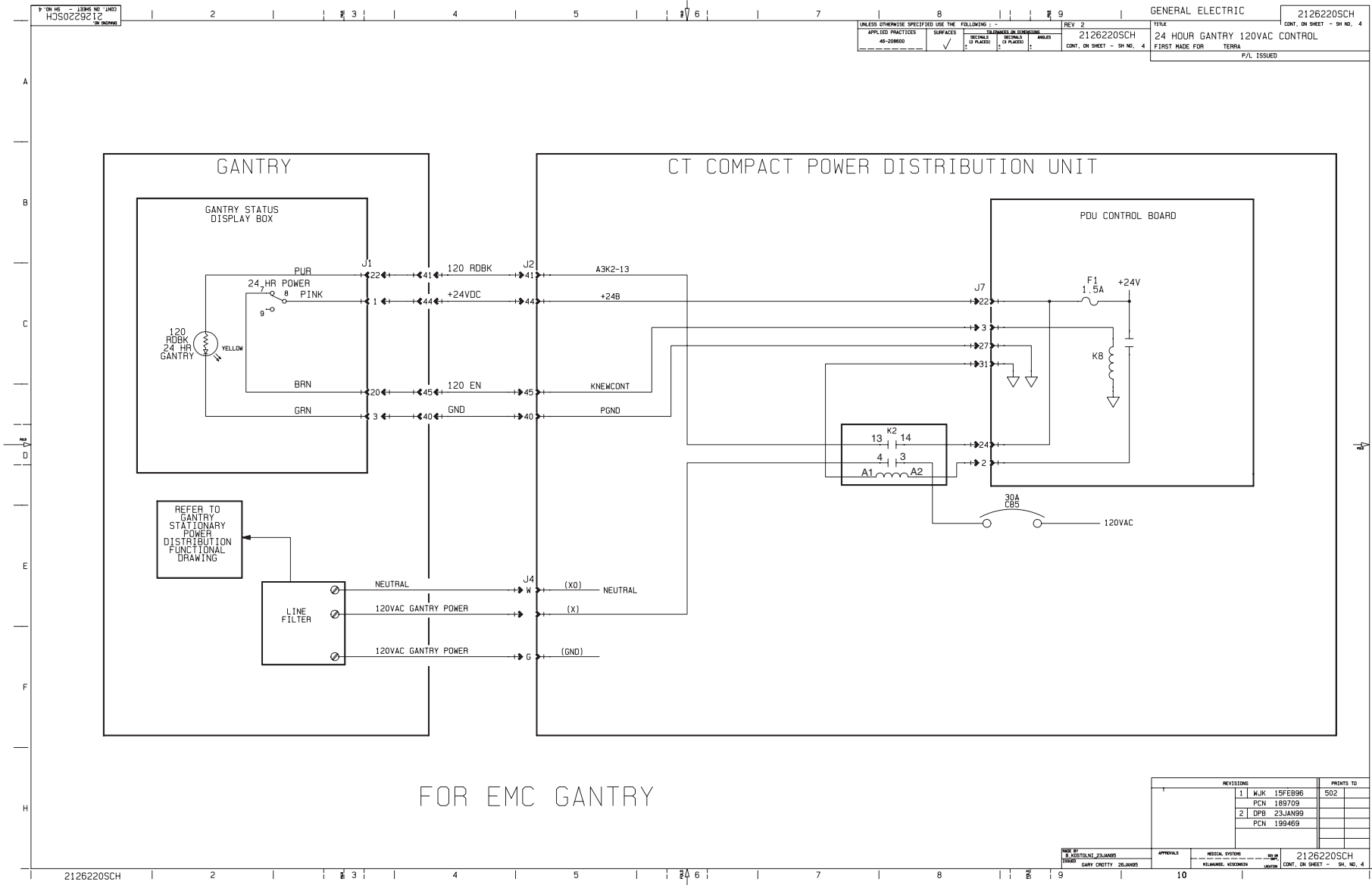


Figure 8-3 2126183 Sheet 3 Rev 2

Section 3.0 - 24 Hour Gantry 120VDC Control - 2126220



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Figure 8-4 2126220 Sheet 4 Rev. 2

Chapter 9

Image Generation

Section 1.0 - Scan Data Save - 2180551

CONSOLE BLOCK DIAGRAM

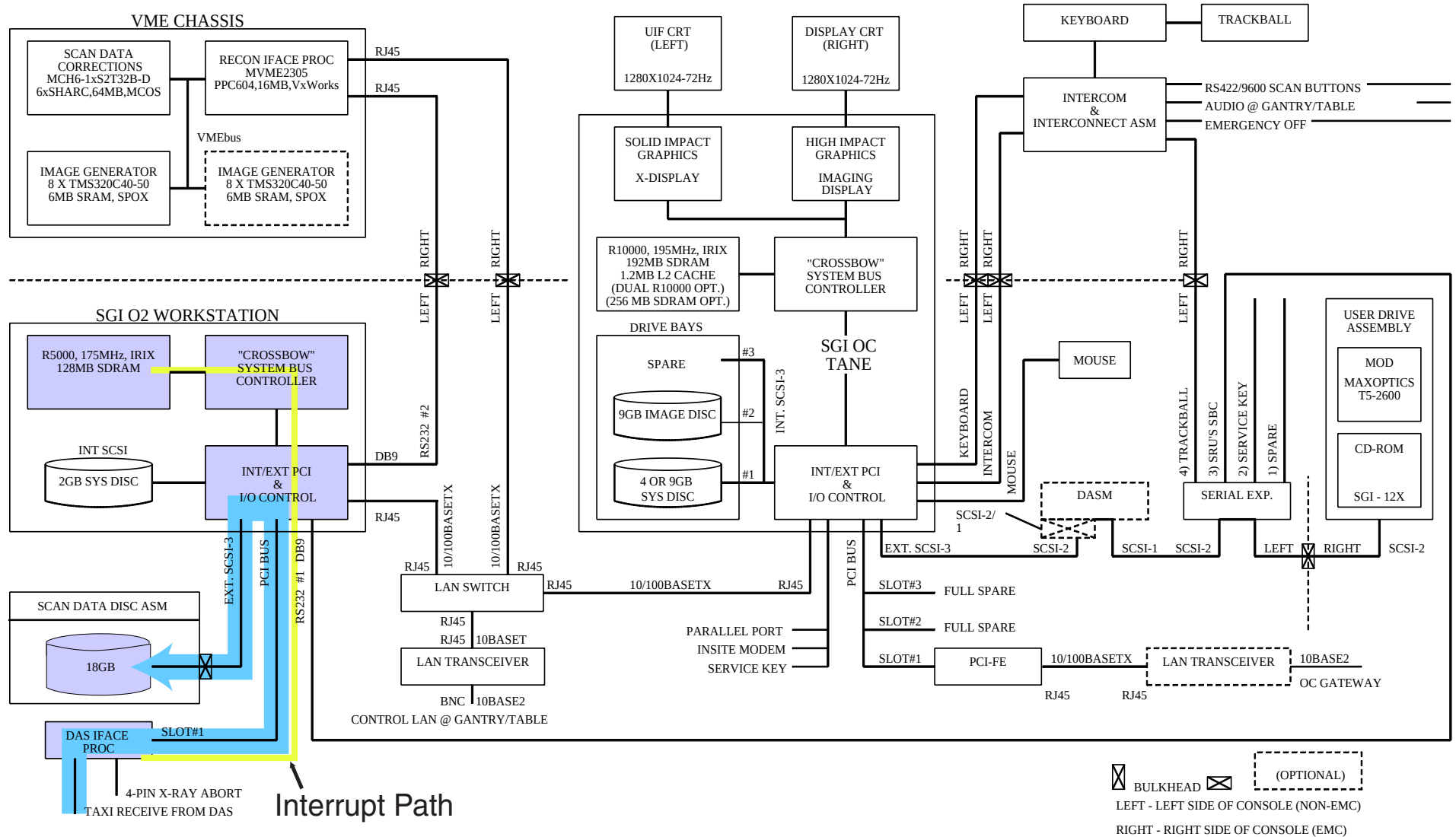


Figure 9-1 2180551 Sheet A

Section 2.0 - Data Restore - 2180551

CONSOLE BLOCK DIAGRAM

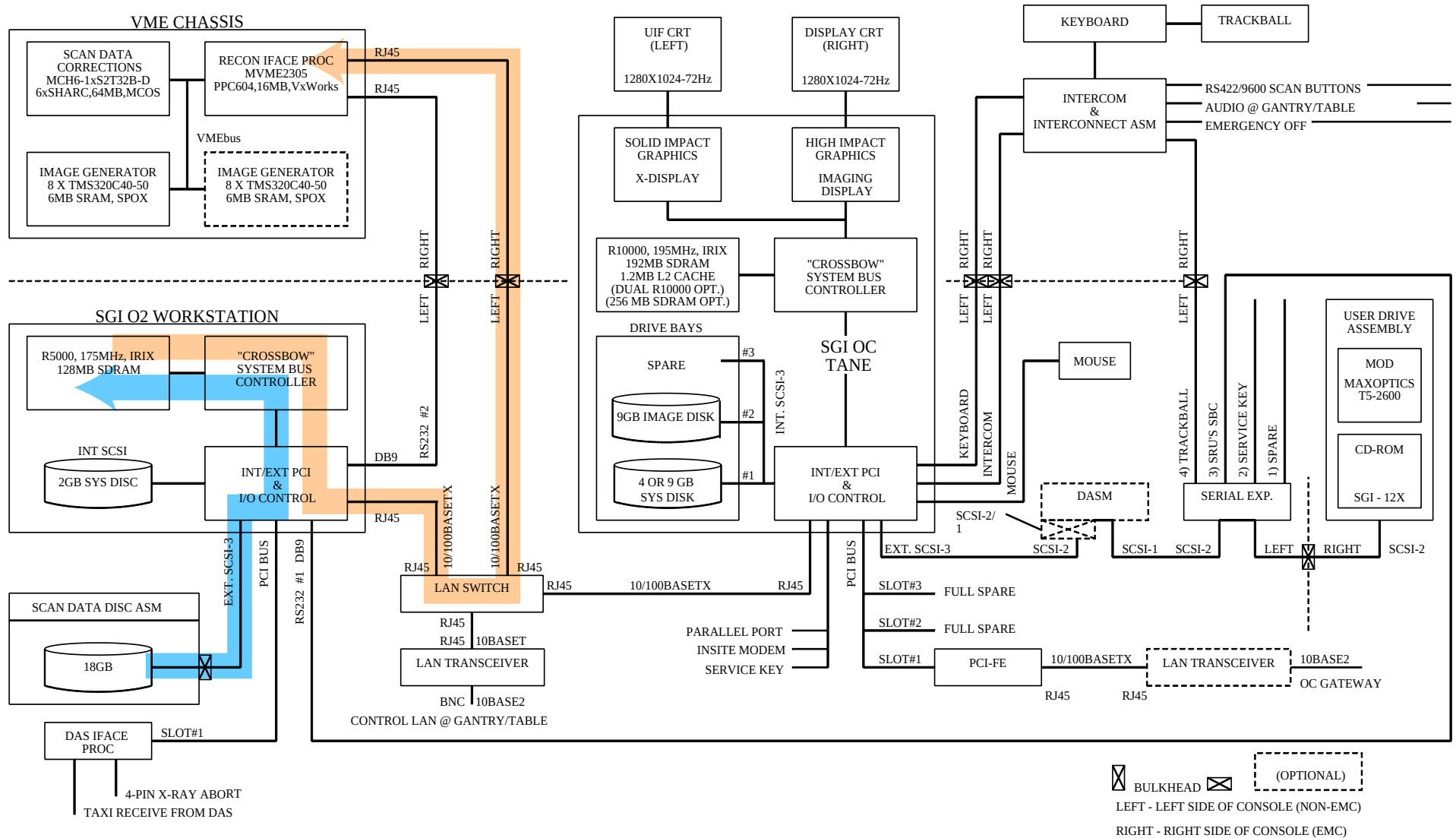


Figure 9-2 2180551 Sheet B

Section 3.0 - Pre-Processing - 2180551

CONSOLE BLOCK DIAGRAM

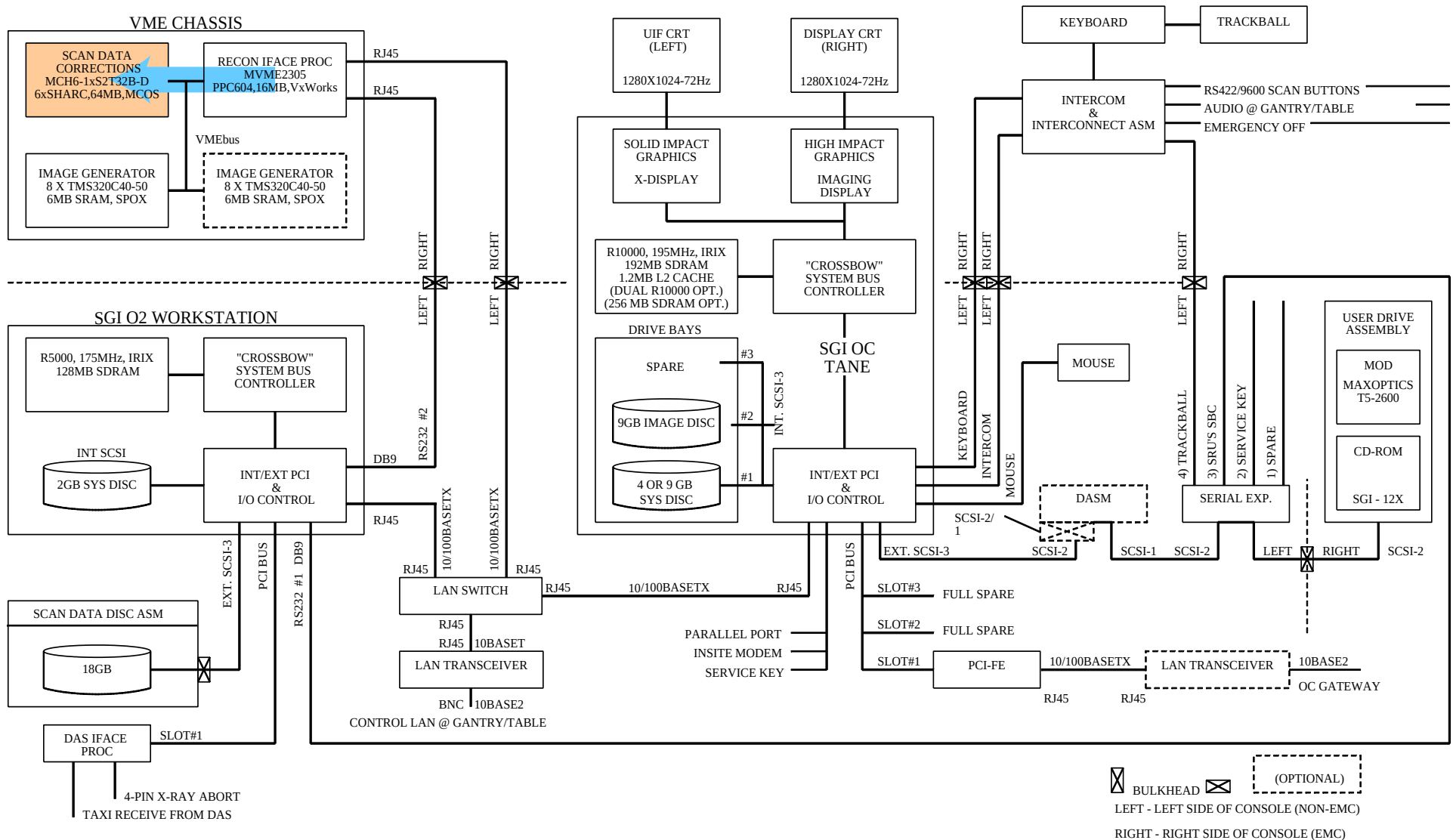


Figure 9-3 2180551 Sheet C

Section 4.0 - Filtered Back Projection - 2180551

CONSOLE BLOCK DIAGRAM

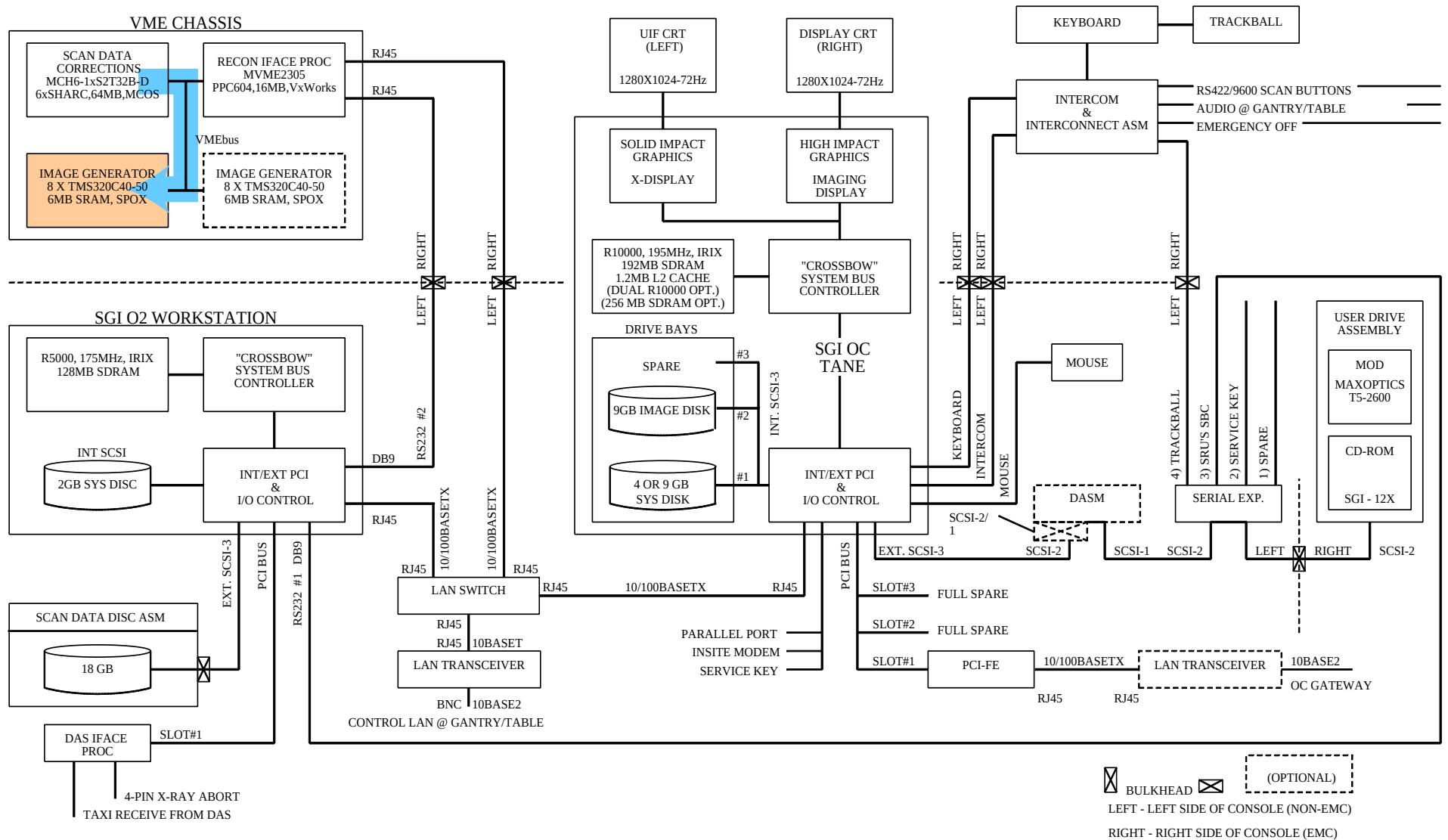


Figure 9-4 2180551 Sheet D

Section 5.0 - Post Processing - 2180551

CONSOLE BLOCK DIAGRAM

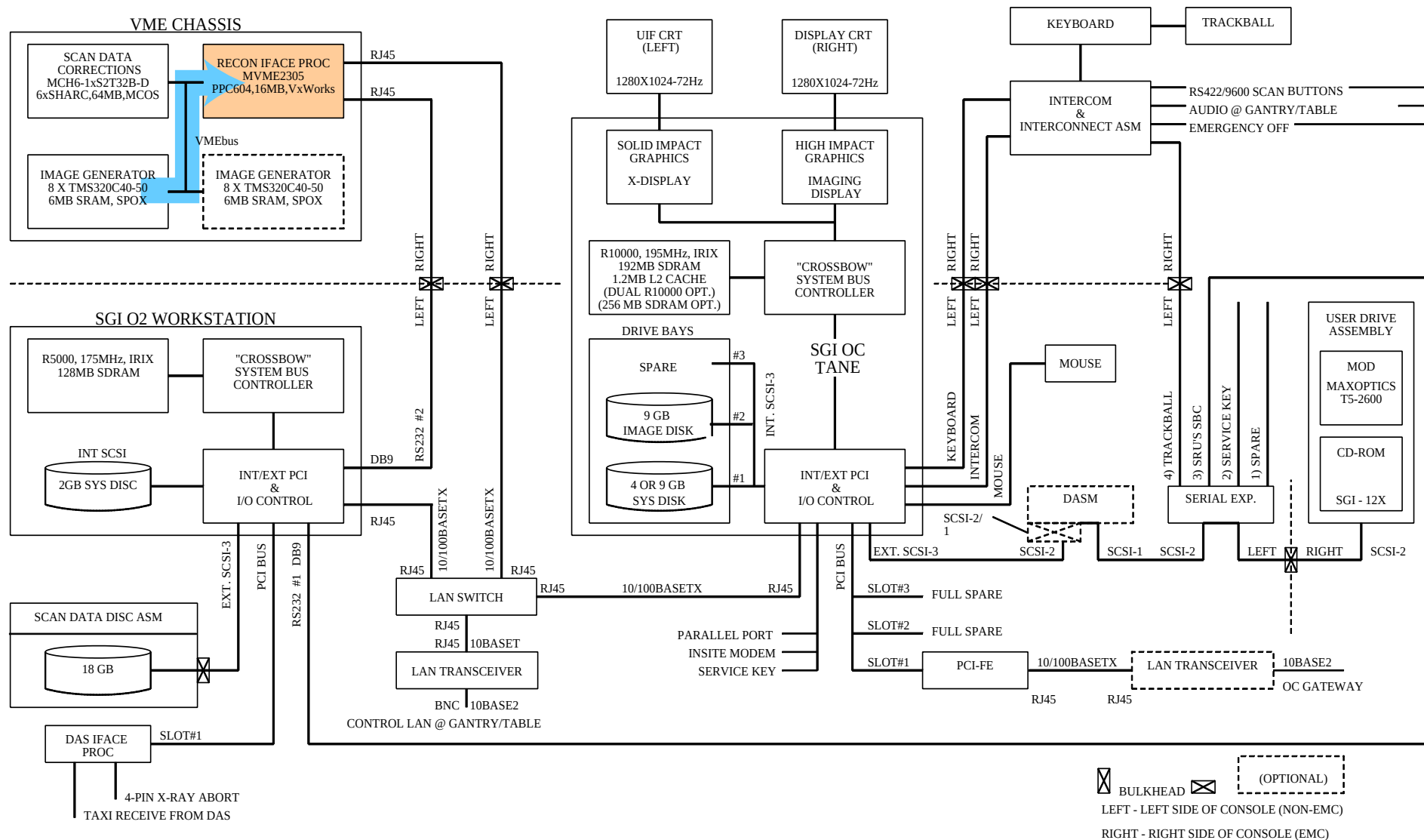


Figure 9-5 2180551 Sheet E

Section 6.0 - Image Transfer - 2180551

CONSOLE BLOCK DIAGRAM

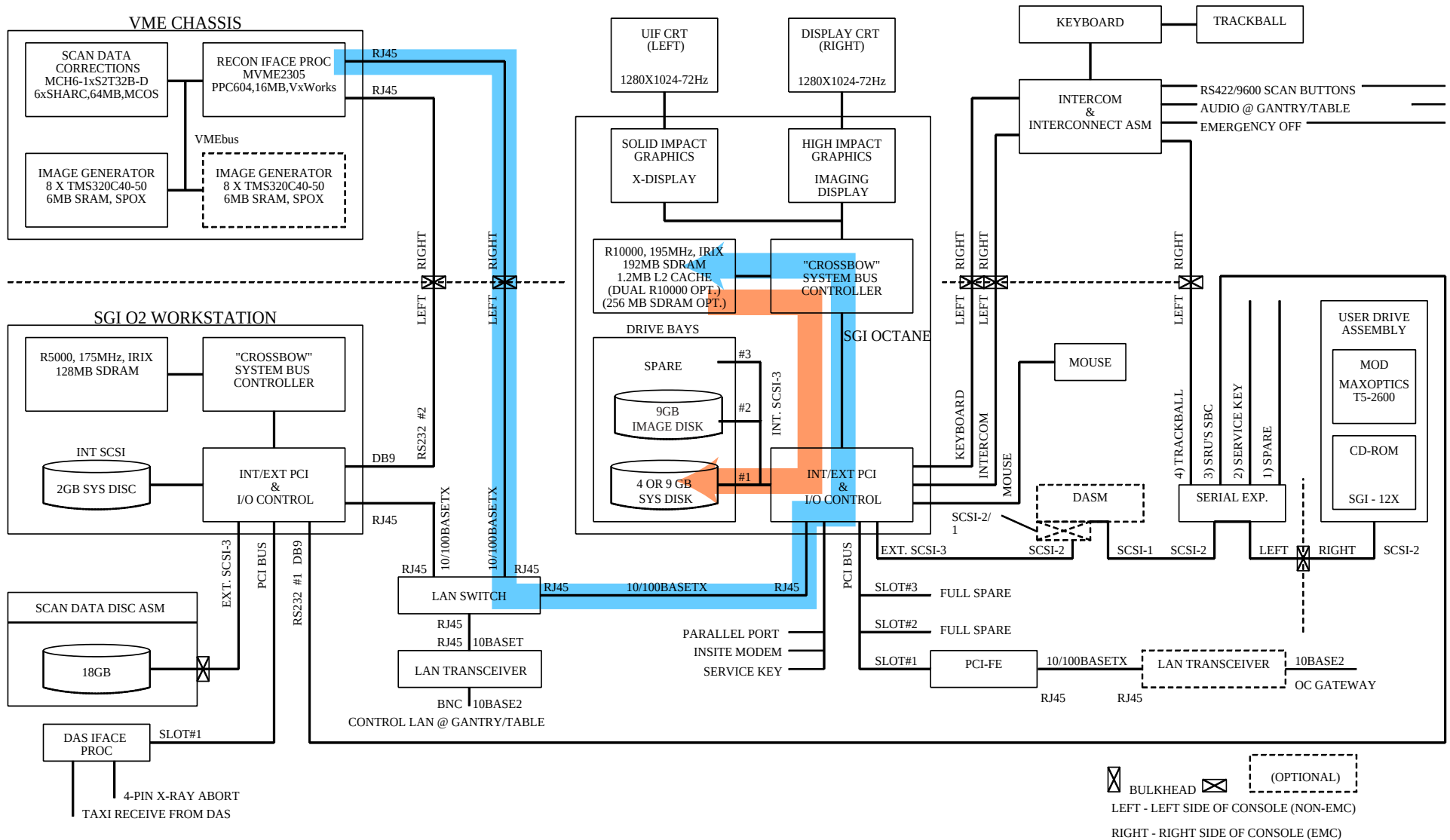


Figure 9-6 2180551 Sheet F



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3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.**

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